Commodity Prices and Sovereign Risk*

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Abstract

Why do global commodity-price increases reduce sovereign risk in some countries but raise it in others? Using data on 41 commodities and 37 countries, we show that trade exposure shapes these differences. We incorporate country-specific commodity baskets into a sovereign-default model in which higher prices raise consumption costs in every country, but affect trade income differently across countries. In exporter-type economies, where higher prices increase net trade income, these gains can offset higher consumption costs; in importer-type economies, trade-income losses reinforce them. The model predicts that spreads fall in exporter-type economies and rise in importer-type economies, consistent with empirical evidence.

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1 Introduction

Global commodity-price fluctuations are a major source of aggregate risk for emerging economies, and sovereign defaults tend to cluster in waves that coincide with these cycles. Booms are usually thought to raise export revenues, improve fiscal positions, and lower sovereign spreads. Yet, these movements lower sovereign risk in some countries and raise it in others. Why does the same global commodity-price shock generate different sovereign-risk responses across countries?

This paper develops a unified framework, built on trade and production data, that links the commodities countries export, import, and consume to sovereign risk over global commodity cycles. Combining empirical evidence with a quantitative sovereign-default model, we show how countries' trade and consumption structures translate the same global price movement into different incentives to repay sovereign debt. Two channels govern this transmission. The first is a price-index channel: higher commodity prices raise the cost of the consumption basket in every country, reducing the purchasing power of available resources. The second is a terms-of-trade channel: higher commodity prices increase earnings from exports but also raise spending on imports, so their effect on trade income depends on the composition of a country's net commodity position. Trade-income gains can offset higher consumption costs, while trade-income losses reinforce them. Exposure to the two channels thus shapes both the direction and the strength of sovereign-risk responses, and explains why the same commodity-price shock affects countries differently.

We develop this argument in three steps. First, we document that the prices of 41 internationally traded commodities share a strong common component, and use unexpected movements in this component as global commodity-price shocks. For 37 countries over 1991–2023, we construct country-specific export and import shocks by combining the global shocks with information on what each country exports and imports and on how strongly those commodities' prices move with this common component. We find that shocks that improve a country's net commodity-trade position lower sovereign spreads. However, when we consider export and import shocks separately, both are associated with higher spreads on average. If net trade income were the only relevant channel, export and import shocks would have opposite effects of similar magnitude, with export shocks lowering spreads.

To understand what drives this pattern, we decompose the export and import shocks into a net component (exports minus imports) and a gross component (their sum). Positive net shocks are associated with stronger external balances and lower government debt,

whereas positive gross shocks are associated with higher spreads, higher debt, and weaker consumption growth. The evidence therefore suggests an additional channel beyond net trade income, and motivates a framework in which commodity prices affect both a country's external resources and the purchasing power of those resources over consumption.

Second, we develop a sovereign-default model with Stone–Geary preferences in which global commodity-price shocks affect both trade income and the cost of consumption. Each country has two exposures, which combine the sensitivity of commodity prices to the global shock with different baskets. Price-index exposure depends on the commodities a country consumes and measures how much the cost of its consumption basket rises when global commodity prices increase. Terms-of-trade exposure depends on the commodities a country exports and imports and determines how its trade income changes. The price-index effect has the same direction in every country, whereas the terms-of-trade effect differs in direction depending on a country's net commodity-trade position. We call an economy exporter-type when a global commodity-price increase raises its export revenues by more than it raises its import costs, and importer-type when the reverse holds. The two exposures shape consumption and repayment incentives, while equilibrium borrowing and default decisions determine sovereign spreads. The terms-of-trade channel is stronger when available resources are low, because a given change in trade income then represents a larger proportional change in consumption.

Third, we calibrate the model to the data. The country-specific parameters come from trade and production data, independently of spread outcomes, and the common parameters are estimated to match average spreads, spread volatility, and the default frequency. This design allows us to assess whether differences in commodity baskets explain why spread responses vary across countries. Terms-of-trade exposure determines the direction of the model-implied spread responses: as global commodity prices rise, spreads fall in all 18 exporter-type economies and rise in all 19 importer-type economies. Although this direction is not targeted in the calibration, the model predicts the empirical direction for most countries with sufficient spread data.

By changing countries' commodity baskets in the model, we can separate the timing of sovereign risk over the commodity cycle from its average level. Whether trade income rises or falls with commodity prices determines whether sovereign risk is higher in booms or in busts. The size of that trade-income response is the main determinant of the average default risk implied by the basket. Assigning the commodity basket of Sri Lanka, an importer-type economy, to Argentina, an exporter-type economy, raises Argentina's model-implied default

frequency, holding all other parameters fixed. A different basket can also shift the peak in sovereign risk from booms to busts without lowering its average level.

These findings imply that global commodity prices alone are not sufficient to assess sovereign risk. The same boom can ease financing conditions in one economy while tightening them in another, and economies exposed in the same direction can differ in their average sovereign risk. Assessing vulnerability requires combining the global commodity cycle with the composition and size of each country's net commodity-trade position. More broadly, the results show how a common global shock spills over unevenly to emerging economies, complementing evidence that global financial conditions transmit across countries through a common factor (Rey 2015, Miranda-Agrippino & Rey 2020). The 2021–22 commodity-price surge illustrates this asymmetry: it raised export revenues in exporter-type economies, while some importer-type economies came under severe debt distress or defaulted. Commodity booms therefore do not provide a uniform signal for emerging-market borrowers: whether a boom raises or lowers sovereign risk depends on the commodity basket.

Related Literature. This paper links the literature on the macroeconomic effects of commodity-price shocks to quantitative models of sovereign default. We show how the same global commodity-price shock can generate opposite sovereign-spread responses across countries, depending jointly on their trade and consumption exposures.

We contribute to the literature on terms-of-trade shocks and business cycles in small open economies (Mendoza 1995, Kose 2002, Schmitt-Grohé & Uribe 2018, Fernández et al. 2018). Our common-factor approach builds on research documenting commodity-price comovement and relating it to global economic fluctuations (Pindyck & Rotemberg 1990, Byrne et al. 2013, West & Wong 2014, Alquist et al. 2020, Delle Chiaie et al. 2022). It also connects to work identifying the underlying drivers of commodity-price fluctuations (Kilian 2009, Kilian & Murphy 2014, Baumeister & Hamilton 2019, Känzig 2021) and documenting the association between global commodity-price cycles and sovereign default waves (Reinhart et al. 2016). Closely related contributions emphasize production and trade structure as determinants of commodity-shock transmission (Kohn et al. 2021), the link between commodity prices and borrowing premia in a net exporter (Drechsel & Tenreyro 2018), and asymmetric effects of import- and export-price shocks (Di Pace et al. 2025). We construct country-specific export and import shocks that combine a common global commodity-price shock with predetermined trade exposures, which allows us to examine how different commodity baskets transmit the same shock to sovereign spreads. Separating export and import shocks reveals a gross component, common to both sides of trade, that is associated with higher spreads,

alongside the lower spreads associated with favorable net shocks. These findings provide evidence of transmission beyond net trade income and motivate the model's distinction between trade income and consumption costs.

Our model builds on the sovereign-default framework of Eaton & Gersovitz (1981), Aguiar & Gopinath (2006), Arellano (2008), and Aguiar & Amador (2014) and contributes to work linking commodity prices to sovereign risk (Min 1998, Cuadra & Sapriza 2006, Hilscher & Nosbusch 2010, Shousha 2016, Lopez-Martin et al. 2019, Roch 2019, Boehm et al. 2021). Relative to this literature, we provide a unified account of heterogeneous sovereign exposure that combines empirical patterns, a structural mechanism, and independently measured country characteristics. Trade and production data determine how a common global shock affects repayment incentives: the terms-of-trade and price-index channels can offset each other in exporter-type economies and reinforce one another in importer-type economies. Because these exposures are measured without using spread data, the direction of the spread response is untargeted, so it provides an out-of-sample test of the mechanism. The counterfactual analysis then separates what determines the timing of sovereign risk over the commodity cycle from what determines its average level.

Another strand of the literature examines how domestic adjustment shapes the transmission of commodity-price shocks. Espino et al. (2025) study sovereign default jointly with distortionary fiscal and monetary policy, showing how policy responses can offset favorable terms-of-trade movements. Esquivel (2025) studies how borrowing and sectoral capital reallocation following commodity windfalls can increase sovereign risk, while Esquivel (2024) develops a related underinvestment channel. Focusing on an endowment economy, we isolate a complementary source of heterogeneity: the trade and consumption baskets through which countries are affected by the same global commodity-price shock. Commodity baskets alone can thus shape repayment incentives, even without domestic policy distortions or sectoral reallocation. These mechanisms relate to the broader literature on resource booms, structural adjustment, and financial vulnerability (Corden & Neary 1982, Benigno & Fornaro 2014, Alberola & Benigno 2017, Hamann et al. 2023, Benguria et al. 2023).

The paper proceeds as follows. Section 2 constructs country-specific export and import shocks and documents how trade exposure shapes sovereign-spread responses to common global commodity-price innovations. Section 3 introduces a sovereign-default model that distinguishes two channels through which commodity prices affect consumption and sovereign risk. Section 4 calibrates the model for the 37 countries in our sample, evaluates its predictions against sovereign-spread data and historical default episodes, and studies counterfactual commodity baskets. Section 5 concludes.

2 Empirical Motivation

2.1 Data Description

Our empirical analysis combines international commodity prices, trade flows, macroeconomic variables, and sovereign spreads for a panel of 37 emerging and developing economies over 1991–2023.[1] The price and trade data begin in 1990, which serves as a pre-sample year for the lagged variables used in the empirical strategy below. We consider 41 internationally traded commodities spanning energy, metals, agriculture, forestry, and livestock; the full list is reported in Table 7 in Appendix A.1. Commodity prices are drawn from the *IMF Primary Commodity Prices Database*. Country-level trade flows are from *UN Comtrade*. Real and nominal GDP are obtained from the *IMF World Economic Outlook*, while exchange rates are from the *IMF Exchange Rate Database*. Inflation, consumption, and government debt are taken from the *Global Macro Database* of Müller et al. (2025). Sovereign risk is measured by EMBI spreads from the *Global Financial Database*, accessed through Finaeon.

2.2 Global Commodity Prices and Country Exposure

International commodity prices exhibit substantial comovement during global booms and downturns (Adler & Sosa 2011, Alquist et al. 2020). Motivated by the broader literature showing that international asset prices and financial conditions contain a strong common component (Rey 2015, Miranda-Agrippino & Rey 2020), we ask how much of the variation in commodity prices can similarly be summarized by a global factor.

We characterize this comovement using a principal component decomposition. Let $P_t(j)$ denote the international price of commodity j, $j \in \{1, \ldots, J\}$. We write:

$$\ln P_t(j) = \mu(j) + \mathbf{a}(j)^\top \mathbf{f}_t + \varepsilon_t(j), \tag{1}$$

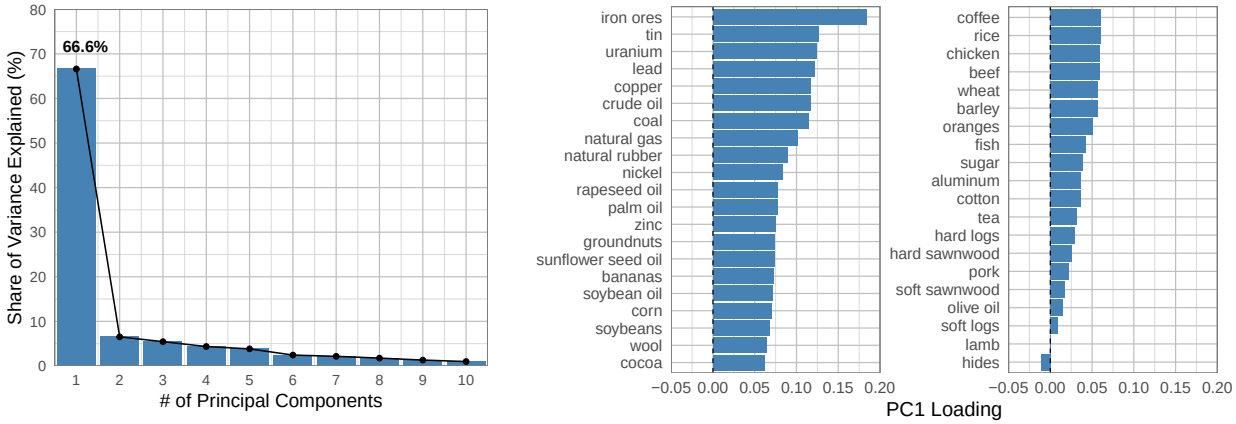
where $\mu(j)$ is the average log price of commodity j, $\mathbf{f}_t$ collects the K common factors in international commodity prices, $\mathbf{a}(j)$ contains the associated factor loadings, and $\varepsilon_t(j)$ captures commodity-specific price variation.[2]

[1] The dataset covers: Argentina, Belize, Brazil, Bulgaria, Chile, China, Colombia, Dominican Republic, Ecuador, Egypt, El Salvador, Gabon, Ghana, Hungary, Indonesia, Iraq, Jamaica, Kazakhstan, Lebanon, Malaysia, Mexico, Pakistan, Panama, Peru, Philippines, Poland, Russia, Serbia, South Africa, South Korea, Sri Lanka, Tunisia, Turkey, Ukraine, Uruguay, Venezuela, and Viet Nam.

[2] The PCA is implemented on standardized log commodity prices: each log-price series is demeaned and scaled to unit variance before extracting the principal components.

Figure 1 shows that the first principal component accounts for most of the variation in commodity prices. Over 1990–2023, it alone explains 66.6% of total variation, more than ten times the 6.5% explained by the second component, and roughly three-quarters of the 86.8% explained jointly by the first five.[3] This degree of comovement is in line with earlier evidence (Byrne et al. 2013, West & Wong 2014, Delle Chiaie et al. 2022). Nearly all commodities load positively on the first component, with similar magnitudes across goods. We therefore interpret it as a broad global commodity-price factor rather than a sector-specific one.

Figure 1: Common Variation in Global Commodity Prices.



(a) Share of variance explained.

(b) Loadings on the first component.

Note: Panel (a) reports the share of total variation in standardized log commodity price levels explained by each principal component for the 41 commodities over 1990–2023. Panel (b) reports the commodity-specific loadings on the first component. Thirty-nine of the 41 loadings are positive, the exceptions being lamb and hides, both livestock-related. The largest loadings span metals, energy, and agriculture, while the smallest are concentrated among commodities with more regionally segmented markets.

Given the dominant role of the first principal component in explaining commodity-price variation, we restrict attention to a single factor ($K = 1$) in the empirical analysis and denote it by f_t, with commodity-specific loadings $a(j)$. To isolate unexpected movements in global commodity prices, we model the factor as an AR(1) process:

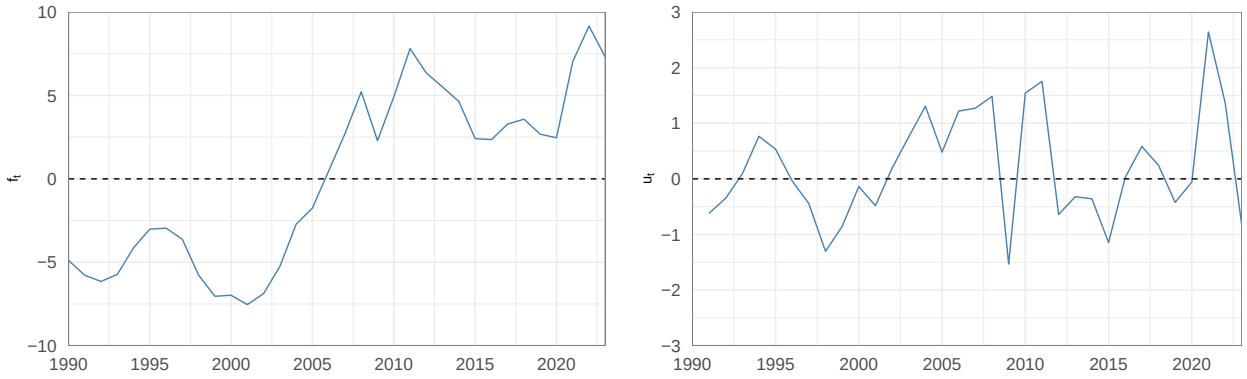
$$f_t = \rho_f f_{t-1} + u_t \tag{2}$$

[3]To assess the role of persistence, Appendix A.2 repeats the analysis on annual log price changes and on commodity-specific AR residuals. Because taking changes and computing AR residuals each use up one observation, these specifications cover 1991–2023. Although the first component explains a smaller share of the variation, it remains the largest, explaining more than three times as much as the second in both cases.

where u_t is the innovation to the global factor. Estimating equation (2) yields a persistence parameter of $\hat{\rho}_f = 0.948$ and an innovation standard deviation of $\hat{\sigma}_u = 1.794$.[4] We standardize the innovation as $\hat{u}_t/\hat{\sigma}_u$, so that a one-unit global shock corresponds to a one-standard-deviation innovation in the commodity-price factor.

Figure 2 plots the estimated global factor together with its standardized innovations. The factor captures broad and persistent movements in global commodity-price conditions, while the innovations isolate the unexpected changes in the factor. In levels, the factor is characterized by a sustained rise in nominal commodity prices from the 1990s through the 2020s. Its largest annual declines occur in 1998, 2009, and 2015, coinciding with the Asian financial crisis, the global financial crisis, and the 2014–15 commodity-price collapse, while the largest increases occur in 2011 and 2021, during the recoveries following the global financial crisis and the pandemic. These episodes largely reflect broad shifts in global demand and macroeconomic conditions rather than idiosyncratic disturbances to individual commodity markets, supporting the interpretation of the first principal component as a common global commodity-price factor. Appendix A.3 further documents positive correlations between factor innovations and indicators of global economic activity, reinforcing the role of global demand in driving common commodity-price movements.

Figure 2: Global Commodity-Price Factor and Innovations.



(a) Global commodity factor.

(b) Innovations to the factor.

Note: Panel (a) plots the first principal component of international commodity prices. Panel (b) reports the innovations u_t obtained from the AR(1) specification in equation (2). Both series are constructed from the 41 commodity prices included in the sample.

[4]The underlying commodity-price data span 1990–2023. Because the global commodity-price shock is defined as the innovation to the estimated factor process, $u_t = f_t - \rho_f f_{t-1}$, the first shock observation is available in 1991. The empirical analysis therefore covers 1991–2023.

Having isolated the global commodity-price innovation, which we treat as our common shock, we next ask why its consequences should vary across the sample. Although the innovation u_t is the same for every country, its incidence depends on what each one trades, since commodity specializations and reliance on exports and imports differ widely. The same innovation can therefore raise export revenues in one country, import costs in another, and both in a third. We exploit this variation in trade structure to construct country-specific measures of exposure to the global innovation.

2.3 Empirical Strategy

Our objective is to evaluate how global commodity-price movements affect sovereign risk, and how this effect depends on a country's commodity trade structure. We begin from the reduced-form specification:

$$\Delta y_{i,t} = \alpha_i + \tau_t + \gamma \Delta P_{i,t} + \eta' X_{i,t} + \varepsilon_{i,t}, \tag{3}$$

where $\Delta y_{i,t}$ is, in our baseline specification, the annual change in country i's EMBI sovereign spread, measured in basis points, $\Delta P_{i,t}$ denotes the commodity-price shock faced by country i, and $X_{i,t}$ is a vector of macroeconomic controls. Rather than proxying this shock with a country-specific price index, we construct it by interacting the global commodity-price innovation with each country's trade exposure.

Because the countries in our sample are small relative to world markets, international commodity prices can be treated as exogenous to conditions in any individual country. The global innovation itself, however, may be correlated with other global developments that also move sovereign spreads: a deterioration in global financial conditions, for example, can depress commodity prices and raise spreads at the same time. A common innovation therefore cannot, on its own, identify its effect on spreads. We thus adopt a shift–share design (Borusyak et al. 2022) in which identification comes from differences across countries in their predetermined exposure to the same innovation.

Recall that u_t is the innovation to the global commodity-price factor and $a(j)$ the loading of commodity j on that factor. A country's exposure to this innovation depends on how heavily its commodity trade is concentrated in goods with large loadings. To distinguish the two sides of the trade account, we measure exposure separately for exports and imports, using lagged trade weights.

Let $X_{i,j,t-1}$ and $M_{i,j,t-1}$ denote exports and imports of commodity j, and define the export and import weights as:

$$\omega^X_{i,j,t-1} = \frac{X_{i,j,t-1}}{\sum_{k=1}^{J}(X_{i,k,t-1} + M_{i,k,t-1})}, \qquad \omega^M_{i,j,t-1} = \frac{M_{i,j,t-1}}{\sum_{k=1}^{J}(X_{i,k,t-1} + M_{i,k,t-1})}. \tag{4}$$

Both weights are shares of total commodity trade. This common denominator preserves the relative importance of exports and imports, which would be lost if the two baskets were normalized separately.

Combining these weights with the factor loadings gives the export and import exposures:[5]

$$\mathcal{E}^X_{i,t-1} = \sum_{j=1}^{J} \omega^X_{i,j,t-1} a(j), \qquad \mathcal{E}^M_{i,t-1} = \sum_{j=1}^{J} \omega^M_{i,j,t-1} a(j), \tag{5}$$

with associated shocks:

$$z^X_{i,t} = \mathcal{E}^X_{i,t-1} u_t, \qquad z^M_{i,t} = \mathcal{E}^M_{i,t-1} u_t. \tag{6}$$

Both shocks are driven by the same global innovation u_t; they differ only in the side of the trade account through which a country is exposed to it. The export shock is larger when the country's export basket is concentrated in commodities that load strongly on the global factor, while the import shock is larger when this is true of its import basket.

Taking the difference, we define net exposure and the net shock as:

$$\mathcal{E}^N_{i,t-1} = \mathcal{E}^X_{i,t-1} - \mathcal{E}^M_{i,t-1}, \qquad z^N_{i,t} = z^X_{i,t} - z^M_{i,t} = \mathcal{E}^N_{i,t-1} u_t. \tag{7}$$

The net shock captures how the global innovation changes the value of commodity exports relative to commodity imports, and therefore summarizes its effect on the country's net external resource position. A positive net shock indicates that the innovation is more favorable through exports than adverse through imports.

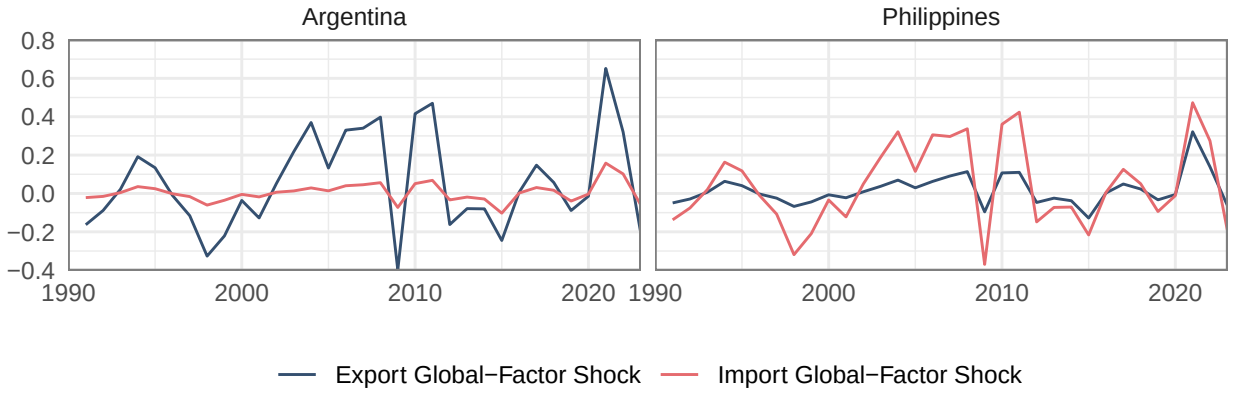
We operationalize $\Delta P_{i,t}$ in equation (3) in two ways. First, we use the net shock $z^N_{i,t}$. Second, we enter the export and import shocks $z^X_{i,t}$ and $z^M_{i,t}$ separately, which allows the two sides of the trade account to have distinct effects. The comparison is informative: if commodity prices affected sovereign risk exclusively through net trade income, the export and import coefficients would be equal in magnitude and opposite in sign. In each case, we control for the corresponding underlying exposures (Borusyak et al. 2022).

[5] Unlike the weights, the exposures are weighted sums of factor loadings rather than shares, and are not restricted to sum to one.

Year fixed effects absorb the effect of the global innovation, and of any other global development, that is common to all countries, so identification relies on cross-country variation in trade exposure.[6] Lagged trade weights rule out contemporaneous adjustment of trade composition to the realized shock,[7] while country fixed effects absorb persistent country characteristics. The identifying assumption is that, conditional on fixed effects and controls, differential exposure is uncorrelated with other country-specific determinants of sovereign spreads. All specifications control for real GDP growth, CPI inflation, and nominal exchange-rate growth, with standard errors clustered at the country level.

Figure 3 illustrates how the same global innovation u_t generates export and import shocks with the same timing and sign but different amplitudes, and how the roles of the two are reversed across countries. The export shock dominates in Argentina, whereas the import shock dominates in the Philippines.

Figure 3: Export and Import Global-Factor Shocks.



Note: The figure reports the country-specific export and import global-factor shocks, $z^X_{i,t}$ and $z^M_{i,t}$, for Argentina and the Philippines. Each combines the innovation u_t with the corresponding lagged trade exposure, $\mathcal{E}^X_{i,t-1}$ or $\mathcal{E}^M_{i,t-1}$.

[6] Section S1.2 of the Supplemental Material replaces year fixed effects with explicit controls for the global commodity-price innovation and global financial conditions, motivated by Sosa-Padilla & Sturzenegger (2023). The estimates closely reproduce the baseline results.

[7] Trade exposure is highly persistent over time. The median year-to-year turnover of the combined export–import commodity exposure vector is 9.0%, with an interquartile range of 6.4–12.1%, while its median cosine similarity with the previous year's vector is 0.989. Persistence remains substantial over longer horizons: at a five-year horizon, median turnover is 18.1% and median cosine similarity is 0.954. These patterns indicate that country-specific commodity exposure is a slow-moving characteristic rather than an object that is extensively reallocated from year to year.

2.4 Results

Baseline sovereign-spread responses. Table 1 reports the effects of global-factor shocks on sovereign spreads. Although the global innovation u_t is standardized, multiplying it by country-specific exposures yields country-level shocks with no natural percentage interpretation. We therefore scale the net, export, and import shocks by the standard deviation of the raw net shock. Using a common divisor preserves the identity $z^N_{i,t} = z^X_{i,t} - z^M_{i,t}$ and implies that the net coefficient corresponds to a one-standard-deviation increase in the raw net shock.

Column (1) shows that a favorable net global-factor shock lowers sovereign spreads: a one-standard-deviation increase in $z^N_{i,t}$ reduces spreads by 36.2 basis points. Thus, when the global commodity-price innovation improves a country's net commodity position, sovereign borrowing costs decline. Column (2) shows that this relationship is concentrated among net exporters, defined by their average net commodity-trade position over 1991–2023.[8] For these countries, the net shock lowers spreads by 77.2 basis points, while the effect is statistically insignificant among the others.

We next ask whether the separate export and import responses are consistent with the view that only net trade income matters. If it were, exports and imports would affect spreads only through $z^N_{i,t} = z^X_{i,t} - z^M_{i,t}$, so their coefficients would be equal in magnitude and opposite in sign. Given the negative net effect in Column (1), export-price gains would lower spreads and import-price increases would raise them. Column (3) does not support this pattern. The import coefficient has the predicted sign (667.1 basis points), but the export coefficient is positive (562.0) rather than negative. Both coefficients being positive points to a component of exposure common to exports and imports, which operates in addition to net trade income.[9] The net effect remains visible in the gap between the two: the export coefficient is smaller than the import coefficient, in line with the negative net coefficient in Column (1).

Column (4) shows that trade orientation shifts the two responses in the direction implied by net trade income. Relative to other countries, net exporters have an export-shock coefficient that is 212.5 basis points lower and an import-shock coefficient that is 110.2 basis points higher. Net exporters thus gain relatively more from export-price increases and

[8] Net commodity-trade positions are highly persistent over time. Although 14 of the 37 countries cross the zero threshold at least once during the sample, the annual classification differs from the time-invariant classification in only 7.9% of country-year observations. This supports treating net-exporter status as a persistent country characteristic. Because it averages over the full sample, the indicator is not predetermined; it is used only to split the sample, not to construct the shocks.

[9] Section S1.1 of the Supplemental Material shows that this pattern is not driven by any single country.

Table 1: Global-Factor Commodity Shocks and Sovereign Spreads.

	Δ Spread			
	(1)	(2)	(3)	(4)
Net Global-Factor Shock	-36.2***	22.6		
	(13.0)	(19.9)		
Net Global-Factor Shock × Net Exporter Dummy		-99.8**		
		(36.9)		
Export Global-Factor Shock			562.0*	1,025.4***
			(288.2)	(334.4)
Import Global-Factor Shock			667.1**	886.2***
			(316.0)	(307.7)
Export Global-Factor Shock × Net Exporter Dummy				-212.5***
				(58.4)
Import Global-Factor Shock × Net Exporter Dummy				110.2**
				(46.6)
Observations	572	572	572	572
R^2	0.31	0.31	0.31	0.32
Country Fixed Effects	✓	✓	✓	✓
Year Fixed Effects	✓	✓	✓	✓

Note: Shocks are scaled by the standard deviation of the raw net shock. Because the same divisor is applied to the net, export, and import shocks, the identity $z^N_{i,t} = z^X_{i,t} - z^M_{i,t}$ is preserved, and the export and import coefficients are not per-standard-deviation effects of their own shocks. The net-exporter indicator is time invariant and equals one when a country's average net commodity-trade position over 1991–2023 is positive. All specifications include country and year fixed effects and controls for real GDP growth, CPI inflation, and nominal exchange-rate growth. Columns (1)–(2) control for net global-factor exposure, while columns (3)–(4) control separately for export and import global-factor exposures. Standard errors, clustered at the country level, are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

lose relatively more from import-price increases. Orientation does not, however, remove the common component: the implied coefficients for net exporters, 812.9 and 996.4 basis points, remain positive.

Column (3) suggests that each coefficient reflects two forces at once. The net position enters the export and import coefficients with opposite signs, whereas the component common to both sides of trade enters them with the same sign. Because both forces are mixed in

every coefficient, neither can be read, or tested, directly. We therefore re-express the export and import shocks as their sum and difference, which gives each force its own coefficient: half the gap between the export and import coefficients is the effect of the net shock, and half their sum is the effect of the gross shock.

Net and gross decomposition. Accordingly, in addition to the net shock defined in equation (7), we define the gross global-factor shock as:

$$z^G_{i,t} = z^X_{i,t} + z^M_{i,t}. \tag{8}$$

The net shock is large when the innovation raises the value of exports more than that of imports. The gross shock is large when both export and import baskets are strongly exposed to the global commodity-price factor, whatever the balance between them. Since $z^X_{i,t} = (z^G_{i,t}+z^N_{i,t})/2$ and $z^M_{i,t} = (z^G_{i,t}-z^N_{i,t})/2$, the net–gross specification is an exact reparameterization of the export–import specification and adds no identifying information.

In Column (1) of Table 2, the net shock lowers spreads by 52.6 basis points, while the gross shock raises them by 614.6 basis points. Because an export shock raises both the net and gross components, its implied coefficient is their sum ($614.6 - 52.6 = 562.0$), exactly reproducing the export coefficient in Table 1. Similarly, an import shock raises the gross component but lowers the net component, so its implied coefficient is their difference ($614.6 - (-52.6) = 667.2$), reproducing the import coefficient up to rounding.

Thus, for an export-price increase, the spread-reducing net component partly offsets the spread-increasing gross component, whereas for an import-price increase the two reinforce one another. The net component behaves as net trade income would predict: favorable net shocks lower spreads. The gross component does not depend on which side of the trade account is hit, and raises spreads in both cases. Exposure to the global innovation therefore has two components with opposite effects on spreads, one tied to the balance between exports and imports and one common to both sides of trade. Column (2) shows that trade orientation operates primarily through the net component. Its interaction with the net-exporter indicator is -161.4 basis points, compared with -51.1 for the gross component. The opposite shifts in export and import responses across trade orientations therefore arise mainly from the net component, while the gross component is only mildly affected by orientation.

When we split the sample by exchange-rate regime, the negative net-exposure association remains statistically significant both under pegs and narrow horizontal bands and under more flexible arrangements. The gross-exposure coefficient remains positive in both subsam-

Table 2: Net and Gross Global-Factor Shocks and Sovereign Spreads.

	Δ Spread	
	(1)	(2)
Net Global-Factor Shock	-52.6***	69.6**
	(18.4)	(26.8)
Gross Global-Factor Shock	614.6**	955.8***
	(301.9)	(320.2)
Net Global-Factor Shock × Net Exporter Dummy		-161.4***
		(47.1)
Gross Global-Factor Shock × Net Exporter Dummy		-51.1**
		(23.8)
Observations	572	572
R^2	0.31	0.32
Country Fixed Effects	✓	✓
Year Fixed Effects	✓	✓

Note: The net and gross global-factor shocks are defined as $z^N_{i,t} = z^X_{i,t} - z^M_{i,t}$ and $z^G_{i,t} = z^X_{i,t} + z^M_{i,t}$. All shocks are scaled as in Table 1. The net-exporter indicator is time invariant and equals one when a country's average net commodity-trade position over 1991–2023 is positive. All specifications include country and year fixed effects, controls for real GDP growth, CPI inflation, and nominal exchange-rate growth, as well as the underlying export and import global-factor exposure measures. Standard errors, clustered at the country level, are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

ples, although it is imprecisely estimated under pegged arrangements (Section S1.3 of the Supplemental Material).

Macroeconomic responses to net and gross shocks. The reduced-form evidence leaves open what the gross component captures. To characterize the two components, we next ask how they relate to external balances, government debt, domestic prices, and consumption (Table 3).

The two components are associated with different macroeconomic responses. A favorable net shock improves external balances and lowers government debt: a one-standard-deviation increase is associated with a 0.76-percentage-point improvement in the current account and a 0.96-percentage-point decline in government debt relative to GDP, and both responses are

Table 3: Macroeconomic Responses to Net and Gross Global-Factor Shocks.

Panel (a): External-Resource Outcomes	Δ CA/GDP		Δ Gov. Debt/GDP	
	(1)	(2)	(3)	(4)
Net Global-Factor Shock	0.756**	-0.063	-0.955*	1.82**
	(0.330)	(0.795)	(0.506)	(0.867)
Gross Global-Factor Shock	8.34	6.96	13.9*	21.6***
	(5.26)	(5.78)	(7.48)	(6.83)
Net Global-Factor Shock × Net Exporter Dummy		1.90*		-3.75**
		(1.04)		(1.56)
Gross Global-Factor Shock × Net Exporter Dummy		-0.336		-1.09
		(0.467)		(0.873)
R^2	0.29	0.30	0.37	0.38
Panel (b): Domestic Prices and Consumption	Inflation		Real Cons. Growth	
	(1)	(2)	(3)	(4)
Net Global-Factor Shock	-0.424	2.43	0.444	-1.52**
	(0.634)	(2.04)	(0.380)	(0.631)
Gross Global-Factor Shock	6.84	16.0	-6.44	-13.4***
	(7.34)	(11.0)	(3.93)	(4.20)
Net Global-Factor Shock × Net Exporter Dummy		-2.61		1.21
		(2.43)		(1.02)
Gross Global-Factor Shock × Net Exporter Dummy		-2.15*		1.96***
		(1.18)		(0.508)
R^2	0.46	0.47	0.58	0.59
Observations	572	572	572	572
Country Fixed Effects	✓	✓	✓	✓
Year Fixed Effects	✓	✓	✓	✓

Note: Panel (a) reports the effects of net and gross global-factor shocks on the annual change in the current account balance and government debt, both expressed relative to GDP. Panel (b) reports effects on CPI inflation and real consumption growth. Real per-capita household consumption growth is calculated using local-currency consumption deflated by the domestic CPI. The series is not converted into US dollars, so exchange-rate movements do not enter it mechanically. All shocks are scaled as in Table 1. All specifications include country and year fixed effects, the corresponding net and gross global-factor exposure measures, and controls for real GDP growth, CPI inflation, and nominal exchange-rate growth. When inflation is the dependent variable, CPI inflation is omitted from the set of controls. Columns (2) and (4) additionally interact the shocks with the net-exporter indicator. Standard errors, clustered at the country level, are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

stronger among net exporters. The gross shock does not improve external balances. It is instead associated with higher government debt and, in the interacted specification, with lower consumption growth. The consumption decline is smaller among net exporters but remains large: the implied response is -11.4 percentage points, compared with -13.4 for other countries, consistent with trade-income gains partly cushioning a loss in consumption that affects all countries. Inflation responses are generally imprecisely estimated.

Taken together, the two components map into two distinct ways in which a commodity-price movement can affect a country. The net component changes the resources a country earns from trade, and it shows up accordingly in external balances and debt, particularly among net exporters. The gross component is common to both sides of trade and therefore cannot be a gain in trade income: it raises spreads and debt and lowers consumption, as if the same global movement also reduced what a given income can buy, whatever the country's trade position. These patterns motivate a structural framework in which a commodity-price increase affects sovereign risk both by changing the resources a country earns from trade and by changing what those resources can finance in consumption.

3 Model

Time is discrete and runs forever. We study a small open economy whose government borrows from risk-neutral international lenders but cannot commit to repay its debt, as in Arellano (2008). Our main departure from the standard sovereign-default framework lies in the source of risk. In place of exogenous endowment shocks, the economy is exposed to global commodity-price shocks that affect both the cost of its consumption basket and the value of its net exports. Default entails temporary exclusion from international credit markets, a loss of consumption, and a direct utility cost.

3.1 Environment

Consider a small open economy that trades a continuum of goods indexed by $j \in [0, 1]$, the continuous counterpart of the commodities in Section 2.[10] The economy takes world prices $P_t(j)$ as given, and these follow the factor structure in equation (1). We abstract from good-specific price variation by setting $\varepsilon_t(j) = 0$, so that commodity prices are driven only by the common factors. Unlike the empirical specification in equation (2), which retains only the

[10] In the quantitative implementation, the continuum is discretized into the 41 commodities of Section 2.

first principal component, the theoretical framework allows for multiple factors following a stationary VAR(1):

$$\mathbf{f}_t = \Phi \mathbf{f}_{t-1} + \mathbf{u}_t, \qquad \mathbf{u}_t \sim \mathcal{N}(\mathbf{0}, \Sigma_u), \tag{9}$$

where Φ governs the persistence and cross-dynamics of the common factors and Σ_u is the covariance matrix of factor innovations.[11]

Households consume all goods according to a Stone–Geary aggregator:

$$c_t = \prod_{j \in [0,1]} [c_t(j) - \bar{c}(j)]^{\lambda(j)}, \qquad \int_0^1 \lambda(j)\, dj = 1,$$

where $\lambda(j)$ denotes the consumption weight and $\bar{c}(j)$ the subsistence requirement of good j. Subsistence requirements allow the economy to be a net buyer of goods whose minimum consumption exceeds domestic endowments. They therefore permit commodity-price increases to reduce trade income, unlike in the corresponding Cobb–Douglas endowment economy (Appendix B.2). Appendix B.1 derives the aggregate environment, and Section S2 of the Supplemental Material considers CES preferences as an alternative to the baseline specification.

Let $e(j)$ denote the endowment of good j. Since subsistence must be met first, what matters is the net endowment $\tilde{e}(j) \equiv e(j) - \bar{c}(j)$: the economy is a net seller of good j when $\tilde{e}(j) > 0$ and a net buyer when $\tilde{e}(j) < 0$. Resources available for spending above subsistence consist of two terms: the net resource inflow from the capital account D_t, defined in equation (14) below, and trade income, defined as the value of net endowments, $\int_0^1 P_t(j)\tilde{e}(j)\, dj$ (Appendix B.1).

Aggregate consumption is proportional to these resources divided by the consumption price index. Global commodity prices therefore affect consumption through two channels: they change the cost of the consumption basket, which is the price-index channel, and they change the resources available to purchase it, which is the terms-of-trade channel. These channels are summarized by the price-index exposure $\boldsymbol{\psi}$ and the terms-of-trade exposure $\boldsymbol{\nu}$, which jointly govern how aggregate consumption responds to global commodity-price movements. We describe each in turn.

[11] In the one-factor case used in the empirical and quantitative analyses, $\mathbf{f}_t$, $\mathbf{a}(j)$, and $\mathbf{u}_t$ reduce to the scalars f_t, $a(j)$, and u_t. Equation (9) then reduces to an AR(1) process of the same form as equation (2), with $\Phi = \rho_f$ and $\Sigma_u = \sigma_u^2$. Section 4.1 describes the estimation of the model's factor process.

Price-index channel. The consumption price index implied by Stone–Geary preferences is:

$$\ln P_t^C \equiv \int_0^1 \lambda(j) \ln P_t(j)\, dj.$$

Using equation (1), it satisfies:

$$\ln P_t^C = \mu^C + \boldsymbol{\psi}^\top \mathbf{f}_t, \qquad \mu^C \equiv \int_0^1 \lambda(j)\mu(j)\, dj, \quad \boldsymbol{\psi} \equiv \int_0^1 \lambda(j)\mathbf{a}(j)\, dj. \tag{10}$$

The price-index exposure $\boldsymbol{\psi}$ is thus the consumption-weighted average of the factor loadings: it is large when household spending is concentrated on goods whose prices load strongly on the global commodity-price factors.

Terms-of-trade channel. Global commodity prices also affect trade income through the value of the net endowment. Linearizing around $\mathbf{f}_t = \mathbf{0}$, where $\bar{P}(j) \equiv e^{\mu(j)}$, gives:

$$\int_0^1 P_t(j)\tilde{e}(j)\, dj \approx \bar{Z} + \boldsymbol{\nu}^\top \mathbf{f}_t, \qquad \bar{Z} \equiv \int_0^1 \bar{P}(j)\tilde{e}(j)\, dj, \quad \boldsymbol{\nu} \equiv \int_0^1 \bar{P}(j)\tilde{e}(j)\mathbf{a}(j)\, dj. \tag{11}$$

The terms-of-trade exposure $\boldsymbol{\nu}$ measures the sensitivity of trade income to global factors. Its sign depends on the commodity composition of net endowments: price increases raise trade income for net sellers and reduce it for net buyers of the goods concerned.

Combining the two channels, aggregate consumption is given by:

$$c_t = \Gamma^c e^{-\mu^C - \boldsymbol{\psi}^\top \mathbf{f}_t} \left(D_t + \bar{Z} + \boldsymbol{\nu}^\top \mathbf{f}_t \right), \tag{12}$$

where $\Gamma^c \equiv \exp\left\{ \int_0^1 \lambda(j) \ln \lambda(j)\, dj \right\}$ is a constant that depends only on the consumption weights.[12] Equation (12), derived in full in Appendix B.1, is exact given the trade-income approximation in equation (11) and is the form used in the numerical solution detailed in Appendix E.

The two channels may offset or reinforce one another. For a factor that raises the consumption price index, positive terms-of-trade exposure partly offsets the increase in consumption costs, whereas negative exposure reinforces it. The relative strength of these effects also depends on resources available for consumption, which are endogenous to the government's borrowing decisions. We formalize these implications in Proposition 1.

[12] It is therefore invariant to prices, common factors, and the resource flow. It scales the level of consumption without affecting the elasticities derived below.

3.2 Government Problem and Market Structure

A benevolent government maximizes expected lifetime utility:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t), \tag{13}$$

where expectations are taken over realizations of the common commodity-price factors, $0 < \beta < 1$ is the discount factor, and $u(c)$ is a strictly increasing and concave period utility function. Aggregate consumption c_t is given by equation (12). The government's problem is subject to the aggregate resource constraint and the bond-market equilibrium conditions.

At the beginning of period t, the state of the economy is $(\mathbf{f}_t, b_t)$, where $\mathbf{f}_t$ denotes the vector of common commodity-price factors and b_t the stock of sovereign assets carried from the previous period. Positive values of b_t correspond to net foreign assets, while negative values correspond to outstanding debt. After observing $\mathbf{f}_t$, the government decides whether to *repay* or *default*. If it repays, it retains market access and chooses next-period assets b_{t+1}.

Resource flows between the country and the rest of the world are:

$$D_t = b_t - q(\mathbf{f}_t, b_{t+1})\, b_{t+1}, \tag{14}$$

where $D_t > 0$ denotes a net resource inflow from the capital account.

Risk-neutral international lenders price bonds competitively to satisfy zero expected profits:

$$q(\mathbf{f}_t, b_{t+1}) = \frac{1 - \delta(\mathbf{f}_t, b_{t+1})}{1 + r^*}, \tag{15}$$

where $\delta(\mathbf{f}_t, b_{t+1})$ is the endogenous probability of default next period, conditional on the current factor state and the government's choice of b_{t+1}.

In default, the economy enters financial autarky and regains access to credit markets with probability $\theta \in (0, 1)$.

3.3 Recursive Competitive Equilibrium

The state of the economy is $(\mathbf{f}, b)$. The equilibrium is characterized by the government's debt policy $b'(\mathbf{f}, b)$, the bond price $q(\mathbf{f}, b')$, and the associated consumption allocation. Let $V^r(\mathbf{f}, b)$ and $V^d(\mathbf{f})$ denote the value functions under repayment and default, respectively:

$$V(\mathbf{f}, b) = \max \left\{ V^r(\mathbf{f}, b), V^d(\mathbf{f}) \right\}. \tag{16}$$

Under repayment, the government retains market access and optimally chooses b'. Let $\pi(\mathbf{f}' \mid \mathbf{f})$ denote the transition density of next period's factor vector $\mathbf{f}'$ conditional on the current state $\mathbf{f}$, implied by the VAR(1) in equation (9). The repayment value satisfies:

$$V^r(\mathbf{f}, b) = \max_{b'} \left\{ u(c) + \beta \int V(\mathbf{f}', b')\, \pi(\mathbf{f}' \mid \mathbf{f})\, d\mathbf{f}' \right\}, \tag{17}$$

where consumption under repayment is:

$$c = \Gamma^c \exp\left(-\mu^C - \boldsymbol{\psi}^\top \mathbf{f}\right) \left[b - q(\mathbf{f}, b')b' + \bar{Z} + \boldsymbol{\nu}^\top \mathbf{f} \right]. \tag{18}$$

The choice of b' is subject to the borrowing limit $b' \geq -\underline{b}$, which rules out Ponzi schemes.

Under default, the government loses access to credit and bears two costs: a fraction $1-\chi$ of consumption and a direct utility cost $\phi(\mathbf{f})$. While in financial autarky, it cannot borrow or lend. The default value satisfies:

$$V^d(\mathbf{f}) = u(c^d) - \phi(\mathbf{f}) + \beta \int \left[\theta V(\mathbf{f}', 0) + (1-\theta) V^d(\mathbf{f}') \right] \pi(\mathbf{f}' \mid \mathbf{f})\, d\mathbf{f}', \tag{19}$$

where consumption under default is:

$$c^d = \chi\, \Gamma^c \exp\left(-\mu^C - \boldsymbol{\psi}^\top \mathbf{f}\right) \left(\bar{Z} + \boldsymbol{\nu}^\top \mathbf{f} \right), \tag{20}$$

and $\chi \in (0, 1)$ is the share of consumption kept in default. The additional utility cost is:

$$\phi(\mathbf{f}) = \max\left\{0,\ \chi_0 + \chi_1 \ln\left(\bar{Z} + \boldsymbol{\nu}^\top \mathbf{f}\right)\right\}, \qquad \chi_0, \chi_1 \geq 0. \tag{21}$$

The two default costs play distinct roles. The first is the consumption loss $1-\chi$, which scales consumption down by the same proportion in every state. Because it acts on consumption, it leaves the elasticity of consumption in default with respect to the factors unchanged: the price-index and terms-of-trade channels both continue to operate. The second is the utility cost $\phi(\mathbf{f})$, which is weakly increasing in trade income and vanishes when $\chi_0 = \chi_1 = 0$. It strengthens repayment incentives when trade income is high, encouraging default to concentrate in low-income states, in the spirit of Arellano (2008). A direct utility cost that rises with tradable income is also used by Bianchi et al. (2023).

Unlike the consumption loss, $\phi(\mathbf{f})$ introduces a dependence on the factors that does not run through consumption. It depends on trade income, $\bar{Z} + \boldsymbol{\nu}^\top \mathbf{f}$, so only the terms-of-trade channel affects it directly; the price-index channel does not. Thus, whenever the interior expression in equation (21) is positive, an increase in factor k raises the utility cost of default if $\nu_k > 0$ and lowers it if $\nu_k < 0$.

Define the repayment and default regions as:

$$\mathcal{A}(b) = \left\{\mathbf{f} \in \mathbb{F} : V^r(\mathbf{f}, b) \geq V^d(\mathbf{f})\right\}, \qquad \mathcal{D}(b) = \left\{\mathbf{f} \in \mathbb{F} : V^r(\mathbf{f}, b) < V^d(\mathbf{f})\right\},$$

where $\mathbb{F} \subseteq \mathbb{R}^K$ denotes the support of the factor vector. The implied probability of default is:

$$\delta(\mathbf{f}, b') = \int_{\mathcal{D}(b')} \pi(\mathbf{f}' \mid \mathbf{f})\, d\mathbf{f}'. \tag{22}$$

When $b' \geq 0$, default is impossible and $\delta(\mathbf{f}, b') = 0$; when $\mathcal{D}(b') = \mathbb{F}$, default occurs with probability one.

Definition 1. *A **recursive competitive equilibrium** consists of functions for (i) consumption $c(\mathbf{f}, b)$ and $c^d(\mathbf{f})$, (ii) debt policy $b'(\mathbf{f}, b)$ and repayment and default sets $\mathcal{A}(b)$ and $\mathcal{D}(b)$, and (iii) bond prices $q(\mathbf{f}, b')$, such that:*

1. *Given policies, $c(\mathbf{f}, b)$ and $c^d(\mathbf{f})$ satisfy the resource constraints (18) and (20).*
2. *Given $q(\mathbf{f}, b')$, the policy $b'(\mathbf{f}, b)$ and sets $\mathcal{A}(b)$ and $\mathcal{D}(b)$ solve the government problem in (16)–(20).*
3. *Bond prices satisfy the zero-profit condition (15).*

Having defined the equilibrium, we now examine how global commodity-price movements affect consumption through the price-index and terms-of-trade channels. The following proposition characterizes the two channels, how their relative strength depends on the resources available for consumption, and when they reinforce or offset one another.

Proposition 1 (Global factors and consumption exposure)**.** *Consider the aggregate consumption equation (12). For a factor k, define* consumption exposure[13]*, holding the resource flow fixed, as:*

$$\Delta_k(\mathbf{f}, b) \equiv \left.\frac{\partial \ln c(\mathbf{f}, b)}{\partial f_k}\right|_D = -\psi_k + \kappa_c(\mathbf{f}, b)\nu_k,$$

where $\kappa_c(\mathbf{f}, b) = 1/[D(\mathbf{f}, b) + \bar{Z} + \boldsymbol{\nu}^\top \mathbf{f}]$ is the inverse of the resources available for consumption, and $D(\mathbf{f}, b)$ is the equilibrium resource flow of equation (14). At the reference state $\mathbf{f} = \mathbf{0}$, this becomes:

$$\Delta_k(\mathbf{0}, b) = -\psi_k + \kappa_c(\mathbf{0}, b)\nu_k, \qquad \kappa_c(\mathbf{0}, b) = \frac{1}{D(\mathbf{0}, b) + \bar{Z}}. \tag{23}$$

[13] Δ_k combines the price-index and terms-of-trade channels defined in Section 3.1. It is an exposure measure, not a distortion: it arises even in the absence of goods-market frictions.

Then:

(i) ***Channel decomposition.*** *The term $-\psi_k$ captures the price-index channel, while $\kappa_c(\mathbf{0}, b)\nu_k$ captures the terms-of-trade channel.*

(ii) ***Sign.*** *For a factor with $\psi_k > 0$, a marginal increase in f_k raises consumption if and only if*

$$\kappa_c(\mathbf{0}, b)\nu_k > \psi_k.$$

Hence a positive terms-of-trade exposure, $\nu_k > 0$, is necessary for consumption to rise.

(iii) ***State dependence.*** *Holding ψ_k and ν_k fixed, the terms-of-trade channel is stronger when the resources available for consumption, $D(\mathbf{0}, b) + \bar{Z}$, are smaller. Thus changes in borrowing conditions can change the magnitude, and when $\nu_k > 0$ potentially the sign, of the consumption response.*

Proof. See Appendix C.1. □

The factor κ_c scales the terms-of-trade channel and depends endogenously on the sovereign's borrowing conditions through the resource flow. A sovereign facing a lower price of debt raises fewer resources from a given issuance, reducing D and increasing κ_c.[14] The strength of this amplification therefore varies with the sovereign's debt position.

The proposition is local: it characterizes the consumption response to a marginal change in factor f_k around the reference state $\mathbf{f} = \mathbf{0}$. Lemma 1 provides sufficient conditions under which the sign of this response is preserved globally: resources available for consumption must remain bounded away from zero and, when $\nu_k > 0$, the terms-of-trade channel must dominate the price-index channel uniformly over the relevant range of states. Appendix C.3 establishes this result.

The consumption exposure characterized in Proposition 1 affects default incentives through the difference between repayment and default values, $G(\mathbf{f}, b) \equiv V^r(\mathbf{f}, b) - V^d(\mathbf{f})$. The sign of Δ_k alone, however, does not determine the response of G, because a factor innovation also moves consumption in default, the state-dependent utility cost of default, and continuation values. Corollary 1, established in Appendix C.2, closes this gap. It gives conditions under

[14] For a given issuance within the borrowing limit of Section 3.2, the price of debt affects D. Feasibility requires $D + \bar{Z} + \boldsymbol{\nu}^\top \mathbf{f} > 0$; Appendix C.3 states results under the assumption that this term remains bounded away from zero.

which G responds to the factor with the sign of Δ_k at the default frontier, so that a positive consumption exposure lowers default probabilities and sovereign spreads, and a negative exposure raises them.

In the one-factor specification used in the empirical and quantitative analyses, $\boldsymbol{\psi}$ and $\boldsymbol{\nu}$ reduce to the scalars ψ and ν, and the consumption exposure becomes $\Delta(f,b) = \kappa_c(f,b)\,\nu - \psi$. The sign of ν classifies economies, as defined in the following remark.

Remark 1 (Exporter- versus importer-type exposure)**.** *In the one-factor specification, ν is the terms-of-trade exposure: its sign reflects how a country's net trade basket is aligned with the global commodity-price factor and determines whether a positive factor innovation raises or lowers trade income. When $\nu > 0$, the country is, on net, a seller of goods that load positively on the factor, so a positive factor innovation raises trade income. We refer to such an economy as exporter-type. Conversely, when $\nu < 0$, a positive factor innovation lowers trade income, and we refer to such an economy as importer-type. This alignment need not coincide with the sign of aggregate net commodity trade, because $\nu = \int_0^1 \bar{P}(j)\tilde{e}(j)a(j)\,dj$ weights each net trade position by its factor loading. Section 4.1 applies this classification to the data through equation (25).*

The classification governs how the two channels interact. Nearly all loadings are positive, so $\psi > 0$: a positive factor innovation raises the cost of the consumption basket. For importer-type economies, the price-index and terms-of-trade channels both reduce consumption, implying $\Delta(f,b) < 0$. For exporter-type economies, the two channels oppose one another: consumption rises following a positive global shock only if the terms-of-trade channel dominates, $\kappa_c(f,b)\,\nu > \psi$. At the reference state $f = 0$, this condition is equivalent to $D(0,b) < D^* \equiv \nu/\psi - \bar{Z}$, so the sign of the consumption response can vary with the debt position.

The model is closely related to the canonical sovereign-default framework of Eaton & Gersovitz (1981) and Arellano (2008). To see this, consider the one-factor specification and shut down the price-index channel by setting $\psi = 0$. Normalizing $\Gamma^c = 1$ and $\mu^C = 0$, consumption under repayment and default becomes $\{c_t = y_t + b_t - q(f_t, b_{t+1})b_{t+1}, c_t^d = \chi y_t\}$, where $y_t \equiv \bar{Z} + \nu f_t$. The global commodity-price factor then enters the sovereign-default problem only through the stochastic income process. Commodity-price movements thus act as trade-income shocks rather than productivity shocks. This distinction is consistent with Kehoe & Ruhl (2008), who show that terms-of-trade changes can affect real income and consumption without a first-order effect on real GDP. Relative to this benchmark, the price-index channel introduces an additional effect of the same global shock on the cost of

consumption, allowing the two channels to offset or reinforce one another, as characterized by Proposition 1.[15] Setting $\psi = \nu = 0$ removes the factor altogether and leaves a deterministic economy in which default does not occur.

The resource flow D_t is the counterpart of the trade balance, which clarifies what $\bar{Z}$ represents. The value of net exports of good j is $P_t(j)[e(j) - c_t(j)]$. Under Stone–Geary preferences, expenditure on good j equals subsistence spending, $P_t(j)\bar{c}(j)$, plus a share $\lambda(j)$ of the resources available above subsistence, $D_t + \int_0^1 P_t(j)\tilde{e}(j)\,dj$. Summing over goods, total net exports equal $-D_t$: as in Arellano (2008), a trade deficit is financed by a net capital inflow. The term $\bar{Z}$ is therefore not the trade balance. It measures net endowments at reference prices and represents the resources available for consumption above subsistence at those prices when $D_t = 0$. Away from that point, D_t is an equilibrium object: the government chooses debt issuance at prices set by lenders, which jointly determine D_t, the default sets, and bond prices.

Consumption exposure, however, is defined at a given resource flow, so it characterizes consumption and not spreads. The sign of Δ therefore does not mechanically determine the response of spreads to the global factor: Section 4.2 shows that Argentina and Uruguay have negative consumption exposure at the reference state, yet their spreads fall with the factor. The type classification, by contrast, matches the sign of the simulated spread response in all 37 countries.

In the one-factor specification, the two channels map into the reduced-form patterns documented in Section 2. The terms-of-trade exposure ν_i depends on a country's net commodity positions and factor loadings, so the net global-factor shock is its closest empirical counterpart. The price-index exposure ψ_i instead depends on the commodity composition of domestic consumption, for which comparable shares across countries are not available at the required level of detail. The gross global-factor shock captures broad exposure common to exports and imports, and its association with higher spreads and weaker consumption is consistent with the role of the price-index channel in the model, without identifying ψ_i itself.[16]

[15] Under CRRA preferences, the normalization of Γ^c only changes the units of consumption and of the associated default-cost parameters. We set $\Gamma^c = 1$ in the quantitative implementation.

[16] Because the export and import weights share a common denominator, $\omega_{ij}^X - \omega_{ij}^M$ is proportional to the commodity-level net trade position. The net shock $z_{i,t}^N$ is therefore, up to normalization, the closest reduced-form counterpart of $\nu_i u_t$, where $\nu_i = \sum_j NX_{ij}a(j)$. By contrast, $\psi_i = \sum_j \lambda_{ij}a(j)$ is built from consumption weights rather than trade shares, so the gross shock does not identify it. The decomposition separates net-trade from common exposure effects, not ν_i and ψ_i country by country.

4 Quantitative Analysis

4.1 Calibration and Estimation Strategy

While the theoretical framework accommodates multiple common factors, the quantitative analysis follows the empirical one in retaining only the first principal component ($K = 1$), whose dominance is documented in Section 2. From this point onward, we therefore suppress vector notation and denote f_t, $a(j)$, u_t, ψ, and ν the scalar counterparts of $\mathbf{f}_t$, $\mathbf{a}(j)$, $\mathbf{u}_t$, $\boldsymbol{\psi}$, and $\boldsymbol{\nu}$, respectively.

The model is solved at a quarterly frequency. The model parameters fall into three groups: country-specific exposures constructed from trade and production data; the process for the global commodity-price factor, estimated from commodity prices; and a small set of parameters common across countries, which are either assigned or estimated. Table 4 summarizes the parameterization.

Country-specific exposures. Country-specific heterogeneity in the quantitative model enters through the price-index exposure ψ_i and the terms-of-trade exposure ν_i, defined in equations (10) and (11). We calibrate these parameters using trade and production data over 1991–2019, valuing commodity flows at fixed reference prices to abstract from historical price fluctuations.[17]

Reference prices are the geometric means of annual international commodity prices over the calibration period.[18] At these prices, we construct average net exports, NX_{ij}, as exports minus imports, and average apparent consumption, C_{ij}, as production minus exports plus imports. Averages use available annual observations over 1991–2019, with negative annual apparent consumption set to zero before averaging. Appendix D.1 describes the data sources, and Appendix D.2 details the construction.

[17] The pre-pandemic calibration avoids incorporating exceptional adjustments in trade and production during 2020–2021 into country-specific exposures, while retaining pandemic-era commodity-price shocks in the empirical analysis. Extending the calibration to 2022–2023 is also constrained by missing copper and aluminum production data. Reference prices are computed over 1991–2019, while factor loadings are estimated using the full 1990–2023 price sample described in Section 2. In Table 6, calibrated country exposures are held fixed and combined with realized commodity-price innovations over 1991–2023.

[18] The model is real and abstracts from nominal exchange rates and the currency composition of sovereign debt. It isolates how global commodity prices affect repayment incentives through trade income and consumption costs. Country exposures are constructed at fixed international reference prices; exchange-rate valuation effects on debt are therefore outside the mechanism studied here. Allowing for debt denominated in different currencies would introduce an additional channel through which commodity-price movements and exchange rates affect debt-service burdens, as emphasized by Ottonello & Perez (2019).

Table 4: Model Parameters.

Parameter	**Symbol**	**Value**
Country-Specific Exposures: Two Illustrative Economies		
Argentina	$(\psi, \tilde{\nu})$	$(0.072,\ 0.069)$
Philippines	$(\psi, \tilde{\nu})$	$(0.070,\ -0.098)$
Global Factor Process: Estimated from Commodity Prices		
Persistence of global factor (quarterly)	ρ_f	0.97
Standard deviation of innovations (quarterly)	σ_u	0.25
Common Parameters: Assigned		
Risk aversion	σ	2
Risk-free rate in percent (quarterly)	r^*	1
Baseline trade income at reference prices (normalization)	$\bar{Z}$	1
Re-entry probability	θ	0.125
Share of consumption kept in default	χ	0.94
Utility cost of default, slope	χ_1	0.17
Common Parameters: Estimated by Simulated Method of Moments		
Discount factor	β	0.94
Utility cost of default, level	χ_0	0.091

Note: The debt grid consists of 201 equally spaced points on $[-0.9, 0.9]$. The borrowing constraint $b' \geq -0.9$ implies a maximum debt level of 0.9, which exceeds mean simulated debt in every country. The factor is discretized on 60 nodes spanning four unconditional standard deviations on each side of its mean. Exposures for all 37 countries are listed in Table 8.

The discrete counterparts of the model's exposure measures are therefore:

$$\bar{Z}_i = \sum_{j=1}^{41} NX_{ij}, \qquad \nu_i = \sum_{j=1}^{41} NX_{ij} a(j), \qquad \lambda_{ij} = \frac{C_{ij}}{\sum_k C_{ik}}, \qquad \psi_i = \sum_{j=1}^{41} \lambda_{ij} a(j). \tag{24}$$

where $a(j)$ is the loading of commodity j on the first component, expressed in log-price units as in equation (1). Thus, ν_i weights net commodity positions by their factor loadings, whereas ψ_i weights the same loadings by domestic consumption shares.

The calibration maps $\bar{P}(j)\tilde{e}_i(j)$ to observed net exports, where $\tilde{e}_i(j) = e_i(j) - \bar{c}_i(j)$. Because endowments and subsistence requirements enter the model only through their difference, they need not be measured separately (Section S2.1 of the Supplemental Material). The mapping omits commodity expenditure above subsistence. This omission is small when commodity consumption consists mostly of subsistence requirements, but it matters when

exports and imports nearly offset: $|\bar{Z}_i|$ then approaches zero and provides an unstable normalization. We therefore impose a floor on the denominator equal to commodity expenditure above subsistence, $R_i = (1-s)\sum_j C_{ij}$, where $s = 0.9$ is the assumed subsistence share of commodity consumption:

$$\tilde{\nu}_i \equiv \frac{\nu_i}{\max\{|\bar{Z}_i|, R_i\}}. \tag{25}$$

The floor binds for three countries whose net commodity trade is close to zero; for the remaining countries, $\tilde{\nu}_i = \nu_i/|\bar{Z}_i|$. The normalization preserves the sign of ν_i and therefore the exporter- versus importer-type classification introduced in Remark 1. Accordingly, we call a country *exporter-type* if $\tilde{\nu}_i > 0$ and *importer-type* if $\tilde{\nu}_i < 0$: a rise in the global factor raises trade income for the former and lowers it for the latter. The classification is determined before solving the model and, as shown in Section 4.3, also coincides with the sign of the simulated spread response.

In the quantitative model, we normalize baseline trade income to $\bar{Z} = 1$, so cross-country differences in the terms-of-trade channel are captured by $\tilde{\nu}_i$. At the reference point, defined by $f_t = 0$ and a zero resource flow, $D = 0$, $\bar{Z}$ represents the resources available for consumption, including income from activities outside the commodity sector. For importer-type economies, a negative $\tilde{\nu}_i$ therefore indicates that higher commodity prices reduce trade income, not that baseline resources are negative. The commodity basket enters the model through ψ_i and $\tilde{\nu}_i$. Under this normalization, $\kappa_c = 1$ at the reference point, yielding the model-implied consumption exposure:

$$\hat{\Delta}_i \equiv -\psi_i + \tilde{\nu}_i. \tag{26}$$

$\hat{\Delta}_i$ determines whether consumption rises or falls in response to a positive global commodity-price shock. Its sign matches that of $\tilde{\nu}_i$ in 35 of the 37 countries: consumption falls in importer-type economies and rises in most exporter-type economies. In the remaining two countries, both exporter-type, the price-index channel outweighs the terms-of-trade gain, so consumption falls despite their positive trade exposure. Appendix D.3 reports $(\psi_i, \tilde{\nu}_i)$ for all 37 countries.

Process of the global commodity-price factor. To estimate the factor process, we extract the first principal component from linearly detrended annual log prices for the 41 commodities over the calibration period, 1991–2019. The first component explains 60.8% of total variation. We normalize the resulting factor to have unit variance and estimate the AR(1) process in equation (2), which at the quarterly frequency of the model implies

$\rho_f = 0.968$ and $\sigma_u = 0.253$.[19] We discretize the resulting process on 60 nodes using the method of Tauchen (1986).

Common parameters. Period utility is of the constant relative risk aversion form, $u(c) = c^{1-\sigma}/(1-\sigma)$, and we set risk aversion to $\sigma = 2$ and the quarterly risk-free rate to $r^* = 1\%$. The re-entry probability is $\theta = 0.125$, implying an expected exclusion period of two years after default, consistent with the evidence for the 1990s reported by Gelos et al. (2011). We set the share of consumption retained in default to $\chi = 0.94$ and the slope of the utility cost of default to $\chi_1 = 0.17$.[20] These parameters are fixed because the targeted moments discipline the overall cost of default but provide limited information about its individual components. Both χ and χ_0 affect the value of default across states, while changes in χ_0 and χ_1 have similar effects on the targeted moments near the reference state.

We estimate the discount factor β and the level of the utility cost of default, χ_0 in equation (21), by simulated method of moments. The estimation has three targets. The first two are spread moments for the 27 countries with more than ten years of EMBI data: the cross-country median of mean annual spreads (336 basis points) and the median within-country standard deviation of annual spreads (170 basis points).[21] The third target is an annual default frequency of 1.8%, based on Moody's sovereign bond defaults over 1993–2019. Country-specific exposures $(\psi_i, \tilde{\nu}_i)$ are constructed from trade and production data without using spread moments, and no moment involving the comovement of spreads with the global factor is targeted.[22] The resulting estimates are $\beta = 0.94$ and $\chi_0 = 0.091$.

The estimated model matches the annual default frequency of 1.8%. It generates a cross-country median of mean annual spreads of 242 basis points, compared with 336 in the data, and a median within-country spread standard deviation of 292 basis points, compared with

[19] These estimates are obtained separately from their empirical counterparts in Section 2, as the factor used in the model differs in sample period, detrending procedure, and normalization.

[20] The proportional loss $1 - \chi$ plays the role of the output cost of default in standard models (Aguiar & Gopinath 2006, Arellano 2008). Setting $\chi = 1$ and re-estimating β and χ_0 gives a similar fit to the targeted moments, and the sign of the spread–factor correlation still matches the classification in every country that prices default risk. Only Uruguay, the exporter-type economy with the smallest terms-of-trade exposure, then borrows up to the debt limit and prices almost no default risk. The results are also unchanged with $\chi = 0.90$.

[21] The ten countries excluded from the estimation are Belize, Gabon, Ghana, Iraq, Jamaica, Kazakhstan, Serbia, Sri Lanka, Tunisia, and Viet Nam, each with eight to ten years of EMBI data.

[22] The classification is robust to alternative exposure constructions. Re-estimating the loadings over 1991–2023 or 1990–2014, the latter excluding the post-boom period, leaves all 37 countries' classifications unchanged. Across these windows, the cross-country correlations between recomputed and baseline exposures are at least 0.98 for ψ_i and 0.96 for $\tilde{\nu}_i$. Constructing consumption weights from import shares rather than apparent consumption yields a similar range of ψ_i but a different country ranking. This change leaves the classification unaffected by construction, since consumption weights enter only ψ_i.

170. The model therefore understates typical spread levels while overstating their volatility. Simulated spreads are low in most states but rise sharply near default. Moreover, risk-neutral lenders require compensation for expected default losses but no additional premium for bearing risk, limiting the model's ability to match observed spread levels. The model generates approximately 70 percent of the cross-country dispersion in mean spreads, a moment not targeted in the estimation: the coefficient of variation is 1.30 in the model and 1.82 in the data.

Given each country's exposures $(\psi_i, \tilde{\nu}_i)$, we normalize $\bar{Z} = 1$ and $\Gamma^c = 1$, hold the price-level constant μ^C common across countries, and solve for the recursive equilibrium using value function iteration and the lenders' zero-profit condition. Appendix E describes the numerical implementation. Section S4 of the Supplemental Material provides further details on factor-process estimation, exposure construction, and the solution algorithm.

4.2 Two Illustrative Economies

We solve the model separately for each of the 37 countries using the country-specific exposures $(\psi_i, \tilde{\nu}_i)$ constructed in Section 4.1. For each calibration, we simulate 500 histories of 10,000 periods. Appendix F reports the corresponding results for the full sample.

To illustrate how the terms-of-trade and price-index channels jointly shape the response of sovereign risk to global commodity-price shocks, we compare Argentina and the Philippines. The two economies have nearly identical price-index exposures, $\psi_{\text{ARG}} = 0.072$ and $\psi_{\text{PHL}} = 0.070$, but sharply different terms-of-trade exposures, $\tilde{\nu}_{\text{ARG}} = 0.069$ and $\tilde{\nu}_{\text{PHL}} = -0.098$, making Argentina exporter-type and the Philippines importer-type according to equation (25). A positive global commodity-price shock therefore raises the cost of consumption by a similar amount in both economies, while increasing trade income in Argentina and reducing it in the Philippines. In Argentina, the trade-income gain largely offsets the higher consumption cost, leaving consumption exposure close to zero ($\hat{\Delta}_{\text{ARG}} = -0.003$). In the Philippines, the loss of trade income compounds the higher consumption cost, producing a substantially negative exposure ($\hat{\Delta}_{\text{PHL}} = -0.168$).

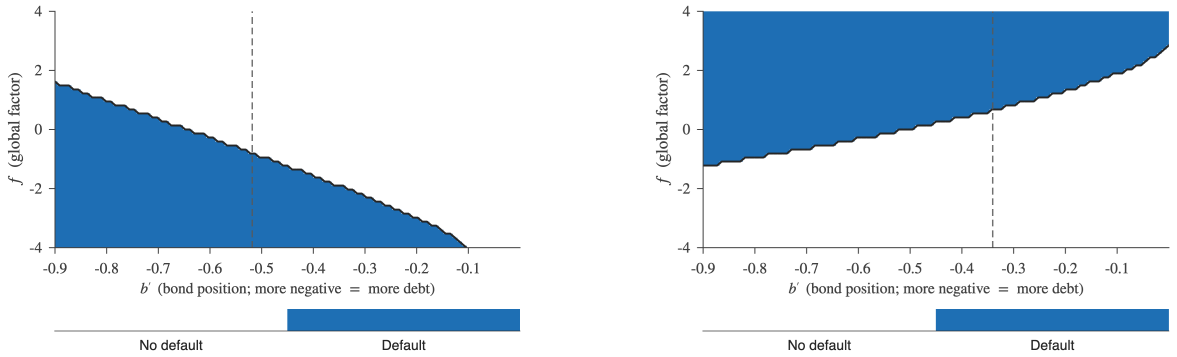
The similarity in price-index exposure does not imply similar consumption baskets. Rather, $\psi_i = \sum_j \lambda_{ij} a(j)$ averages commodity loadings using consumption weights that sum to one, allowing differences across commodities to offset.[23] With relatively similar loadings across commodities, this averaging limits the dispersion of ψ_i, which ranges from 0.053 to

[23] Argentina's apparent consumption is concentrated in soybeans, crude oil, beef, and natural gas, while the Philippines' is concentrated in rice, bananas, crude oil, and pork. The contributions of individual

0.105 across the 37 countries. By contrast, the net-export weights entering $\tilde{\nu}_i$ can be positive or negative and need not sum to one, allowing terms-of-trade exposure to vary much more widely across countries.

Figure 4 shows the equilibrium default sets. Greater indebtedness raises default incentives in both economies, but the global factor shifts the default region in opposite directions. In Argentina, default occurs at low realizations of the factor, whereas in the Philippines it occurs at high realizations. Higher global commodity prices thus reduce sovereign risk in Argentina and increase it in the Philippines.

Figure 4: Default Sets.



(a) Argentina.

(b) Philippines.

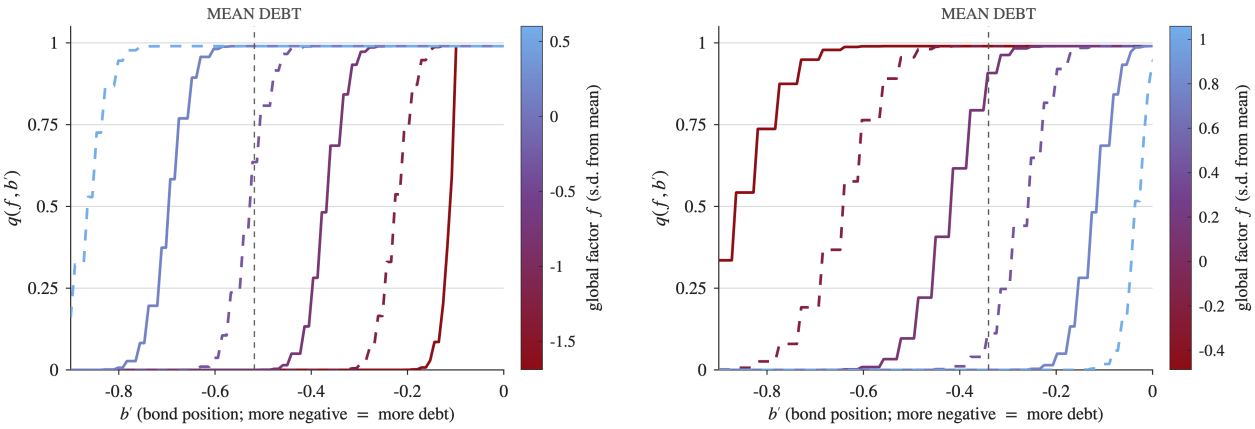
Note: Each panel shows the combinations of debt b and global factor f for which default is optimal (shaded region). Default sets for all 37 countries in the sample are reported in Section S5.1 of the Supplemental Material.

Argentina illustrates why $\hat{\Delta}_i$ alone does not determine the response of sovereign spreads. Its consumption exposure is slightly negative ($\hat{\Delta}_{\text{ARG}} = -0.003$), yet its spreads fall with the global factor in the quantitative model. Two equilibrium forces explain this result. First, $\hat{\Delta}_i$ is evaluated at $D = 0$, where $\kappa_c = 1$. A government that services debt has $D < 0$, so the resources available for consumption fall below trade income, $\kappa_c > 1$, and the terms-of-trade channel is amplified. Second, the utility cost of default rises with trade income (equation (21)), which makes repayment more attractive when global commodity prices rise. Uruguay is the only other exporter-type economy with $\hat{\Delta}_i < 0$ whose spreads nevertheless fall with the global factor.

commodities to $\psi_{\text{ARG}} - \psi_{\text{PHL}}$ differ substantially: soybeans contribute $+0.013$, rice -0.013, and bananas -0.012. The absolute contributions sum to more than thirty times the net difference of 0.002.

Figure 5 shows the corresponding bond-price schedules. Consistent with the default sets, higher realizations of the global factor raise bond prices in Argentina but lower them in the Philippines. In both economies, bond prices decline with indebtedness. The slope of the schedule also depends on the magnitude of terms-of-trade exposure, $|\tilde{\nu}_i|$. When $|\tilde{\nu}_i|$ is large, default incentives respond strongly to the factor, so different realizations of f generate widely separated default thresholds and the schedule is more gradual in b'. When $|\tilde{\nu}_i|$ is small, these thresholds are tightly clustered and the schedule approaches a step function (Supplemental Material, Section S5.2).

Figure 5: Bond Price Schedules.



(a) Argentina. (b) Philippines.

Note: Each panel shows the equilibrium bond-price schedule $q(f, b')$ as a function of next-period bond holdings b' for different realizations of the global factor f. Each line corresponds to one value of f, from dark red (low) to light blue (high), as shown in the color bar; the dashed vertical line marks mean simulated debt. Bond-price schedules for all 37 countries are reported in Section S5.2 of the Supplemental Material.

Table 5 summarizes the simulated moments for these two economies, while Table 10 in Appendix F.1 reports them for all 37 countries. Holding other characteristics fixed, the Philippines' commodity exposure generates greater default risk than Argentina's, and the sign of the spread–factor covariance is reversed. In Argentina, the covariance between spreads and the global factor is negative, -0.32. Consumption is relatively stable, with a standard deviation of 0.029 and a small covariance with the factor of 0.002. The economy sustains average debt of 0.52 and a mean spread of 0.56%.[24] In the Philippines, the spread–factor

[24]Mean consumption differs slightly across the two economies despite the normalization $\bar{Z}_i = 1$ in both cases. Almost all of the difference, 0.013, reflects differences in debt-service flows, while the price deflator accounts for only 0.0002. Argentina carries more debt and therefore devotes a larger share of its resources to debt service.

covariance is positive, 3.23, consumption is substantially more volatile, with a standard deviation of 0.180, and its covariance with the factor is -0.177. Average debt is lower, at 0.34, and the mean spread is higher, at 2.83%.

This comparison isolates the contribution of commodity exposure to sovereign risk, not overall country risk. Observed spreads are higher in Argentina than in the Philippines, reflecting additional sources of heterogeneity, including output fluctuations, institutions, and default histories, that the model does not capture.

Table 5: Simulation Moments for the Two Illustrative Countries.

	Annual Spread (%)			Debt			Consumption		
Country	Mean	Std	Cov w/ F	Mean	Std	Cov w/ F	Mean	Std	Cov w/ F
Argentina	0.560	0.657	-0.320	0.518	0.152	0.140	0.992	0.029	0.002
Philippines	2.834	5.787	3.233	0.340	0.199	-0.186	1.005	0.180	-0.177

Note: Each row reports simulation moments for one country over 500 runs of 10,000 periods. Annual Spread is in percent. Debt and Consumption are in model units with $\bar{Z}_i \equiv 1$. Cov with F is the sample covariance with the global factor. The global factor itself has Mean ≈ -0.002 and Std ≈ 1.007 across the two calibrations.

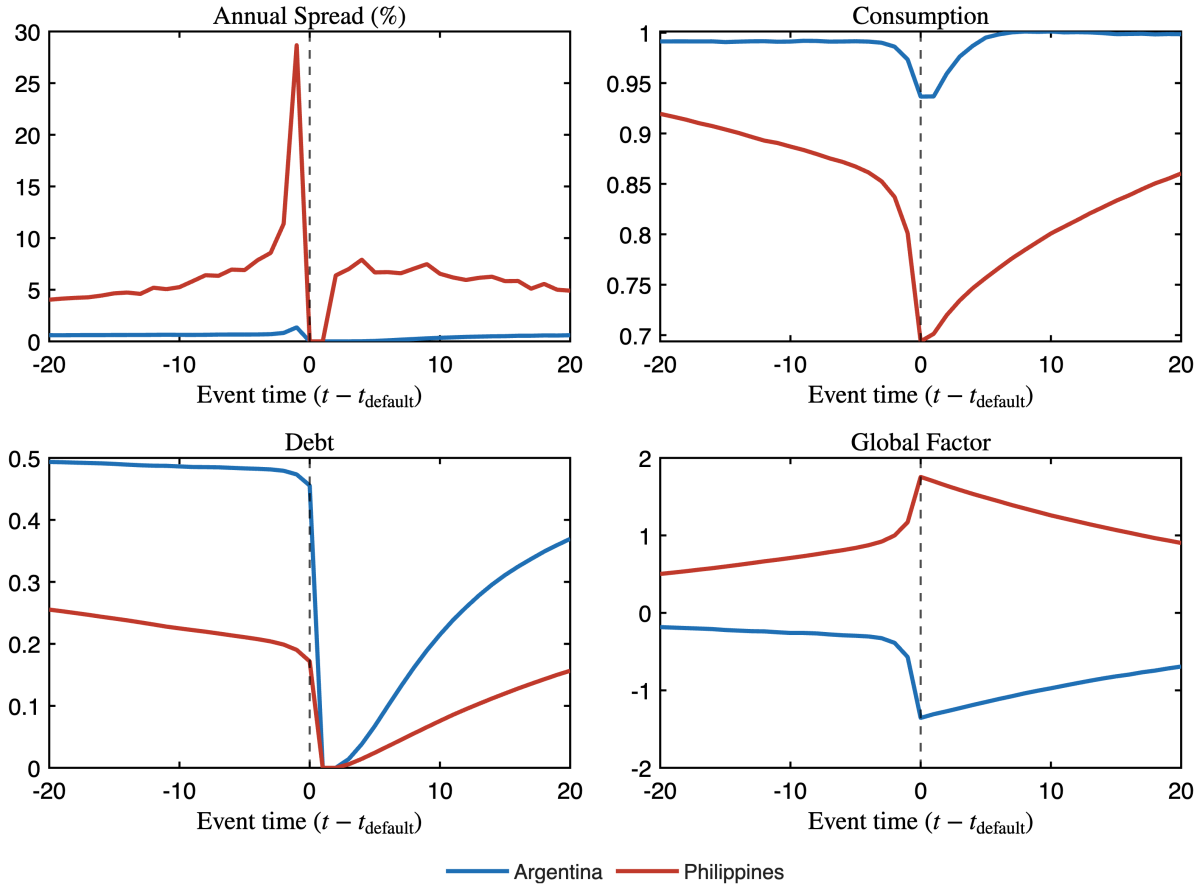
Figure 6 illustrates how these differences appear around default episodes. In Argentina, defaults coincide with sharp falls in the global factor, whereas in the Philippines they coincide with sharp rises: at the default date, the factor is about 1.4 standard deviations below its mean in Argentina and about 1.7 above it in the Philippines. Spreads rise sharply before default in both economies, but the deterioration is more persistent in the Philippines because the terms-of-trade and price-index channels reinforce one another.

Taken together, the two calibrations show how differences in commodity exposure can reverse the equilibrium response of sovereign risk to the same global shock. Because the two economies have nearly identical price-index exposures, the reversal comes from the sign of terms-of-trade exposure. Section 4.3 shows that this sign separates the response of spreads across all 37 countries.

4.3 Cross-Country Heterogeneity

The sign of terms-of-trade exposure $\tilde{\nu}_i$ (Remark 1 and equation (25)) predicts the direction of the spread response in every country. In the simulations, the covariance between spreads and the global factor is negative in all 18 exporter-type countries and positive in all 19 importer-

Figure 6: Dynamics Around Default Events.



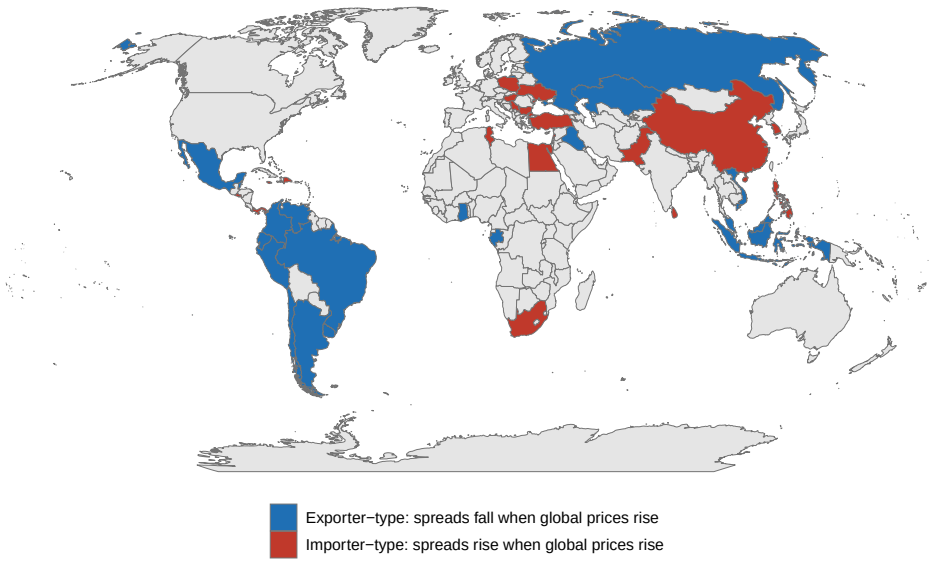
Note: Each line traces the average dynamics of a key variable around a default event for one country. Argentina is shown in blue and the Philippines in red. Default events are drawn from 500 simulations of 10,000 periods per country. Figure 14 in Appendix F.2 reports the corresponding dynamics for all 37 countries in the sample.

type countries (Table 10 in Appendix F.1). This correspondence is robust to changes in the common parameters whenever the model generates default risk.[25] Figure 7 maps the direction of this response across the sample. Higher commodity prices raise trade income in exporter-type economies, which strengthens repayment incentives and lowers spreads. In importer-type economies, they reduce trade income while raising the cost of consumption, which increases default risk and widens spreads. The classification also coincides with ob-

[25] We re-solve the model for all 37 countries under six alternative parameter vectors, each changing the baseline in one direction: $\theta = 0.20$, $\chi_1 = 0.40$, $\beta = 0.92$, $\chi = 0.90$, $\chi_0 = 0.12$, and $(\theta, \chi_0, \chi_1) = (0.20, 0.20, 0.40)$. These vectors generate median spreads ranging from 57 to about 15,000 basis points. In every case, the sign of the spread–factor correlation matches the country classification in all countries that price default risk (35 to 37 of the 37).

served commodity trade positions: the sign of $\tilde{\nu}_i$ matches that of aggregate net commodity trade in all 37 countries.

Figure 7: Simulated Response of Sovereign Spreads to the Global Commodity Factor.



Note: Colors summarize the model's predictions based on country-specific exposure parameters $(\psi_i, \tilde{\nu}_i)$. Blue denotes exporter-type countries ($\tilde{\nu}_i > 0$), for which a global commodity expansion reduces default risk, while red denotes importer-type countries ($\tilde{\nu}_i < 0$), for which it increases default risk. Gray indicates countries outside the sample. Country-level values are listed in Appendix D.3.

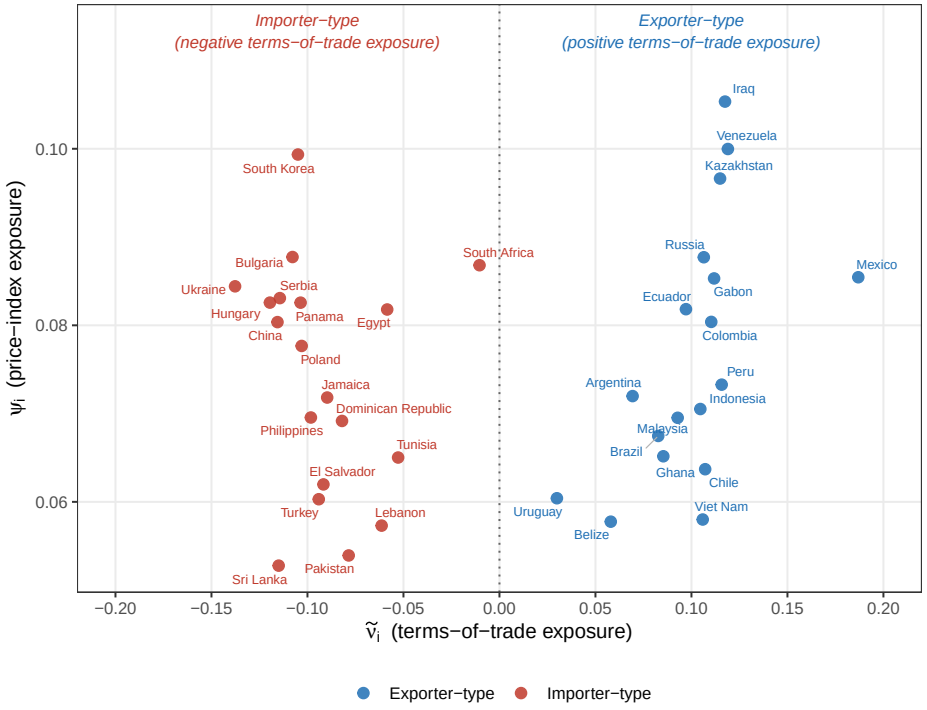
Figure 8 shows the calibrated exposure pairs $(\tilde{\nu}_i, \psi_i)$. Cross-country heterogeneity is concentrated in terms-of-trade exposure: ψ_i is tightly clustered, whereas $\tilde{\nu}_i$ varies widely, with zero as the threshold between exporter- and importer-type economies. The largest positive exposures are in Mexico, Venezuela, Iraq, Peru, and Kazakhstan; the largest negative exposures in absolute value are in Ukraine, Hungary, China, Sri Lanka, and Serbia.

Consumption rises with the global factor in most exporter-type countries, where the terms-of-trade channel dominates the price-index channel. Argentina and Uruguay are the exceptions: at $\kappa_c = 1$, their terms-of-trade exposure is too small to offset the price-index channel. As discussed in Section 4.2, their spreads nevertheless fall with the factor because of equilibrium forces.

The degree of offset between the channels is also associated with differences in borrowing outcomes. The 17 countries with $|\hat{\Delta}_i| < 0.05$, comprising every exporter-type country except Mexico, carry more debt than the remaining countries (0.43 versus 0.37 in units of trade income) and borrow at lower average spreads (1.8% versus 4.0%). This comparison largely

reflects the distinction between economies in which the channels offset and those in which they reinforce one another. Uruguay, which has the smallest positive terms-of-trade exposure, also displays the weakest spread cyclicality (-0.02).

Figure 8: Cross-Country Exposure Parameters.



Note: Each point shows a country's calibrated $(\tilde{\nu}_i, \psi_i)$. Colors indicate whether a country is exporter-type ($\tilde{\nu}_i > 0$) or importer-type ($\tilde{\nu}_i < 0$). This classification also matches, for every country, the sign of the simulated covariance between sovereign spreads and the global factor. The vertical dotted line marks $\tilde{\nu}_i = 0$.

Taken together, the cross-country results show that the two channels play different roles. Heterogeneity in the direction and strength of sovereign-risk responses comes primarily from terms-of-trade exposure, which varies widely in both sign and magnitude. The price-index channel is similar across countries: ψ_i is positive in all 37, so a rise in global commodity prices raises the cost of consumption everywhere. It therefore does not distinguish countries, but it shapes how the global factor affects consumption, offsetting a substantial part of the trade-income gain in exporter-type economies and compounding the loss in importer-type economies. Section 4.4 asks whether this model-implied heterogeneity, together with the common increase in consumption costs, is reflected in the empirical responses to global commodity-price shocks.

4.4 Model Validation

We assess the model along three untargeted dimensions: the cross-country cyclicality of sovereign spreads, the timing of selected historical default episodes, and the relationship between the calibrated exposure parameters and the reduced-form responses documented in Section 2.

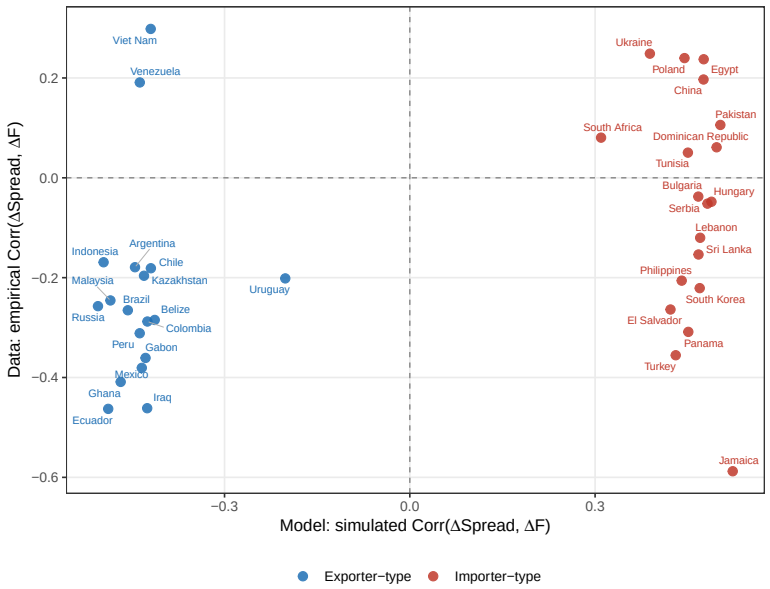
Figure 9 compares simulated and empirical spread cyclicality across countries. The calibration targets the level and volatility of spreads and the frequency of default, and includes no moment involving comovement between spreads and the global factor. In both the model and the data, cyclicality is measured as the correlation between annual changes in spreads and annual changes in the global factor. The model predicts the empirical sign of spread cyclicality for 18 of the 27 countries with more than ten years of EMBI data, and the rank correlation between simulated and empirical cyclicality is 0.49. Exporter-type economies are concentrated in the lower-left quadrant, where both the model and the data imply that spreads fall when global commodity prices rise. Importer-type economies are more dispersed in the data, suggesting a greater role for other global and domestic drivers of sovereign spreads.

Across all 37 countries, the average absolute correlation is 0.45 in the model and 0.24 in the data. The stronger simulated comovement is consistent with the model's focus on commodity-price uncertainty, whereas observed spreads respond to many other shocks.[26] The exercise therefore speaks primarily to cross-country differences in the direction of sovereign-risk responses.

Figure 10 provides a complementary historical illustration. The exporter-type defaults, Russia in 1998, Ecuador in 1999, and Argentina in 2001, occur when the global commodity factor is low. Among importer-type economies, Sri Lanka's default in 2022 occurs near the peak of the factor, and Lebanon's in 2020 occurs when the factor is above its sample average, shortly before the 2021 surge. The small number of episodes limits what they can show, and their timing also reflects fiscal, political, and financial forces outside the model. The exporter-type defaults also fall early in the sample, when the factor is generally lower, and the importer-type defaults late, when it is generally higher, so part of the alignment may reflect the factor's upward trend. We therefore treat these episodes as illustrations and rely on the cross-sectional evidence in Figure 9 and Table 6.

[26] Among the 27 countries with more than ten years of EMBI data, the squared correlation between annual changes in spreads and the global factor has a median of 5%. It exceeds 10% in only three countries, reaching a maximum of 21% for Ecuador.

Figure 9: Model Prediction vs. Data: Spread Cyclicality.



Note: The horizontal axis reports the simulated correlation between annual changes in spreads and annual changes in the global factor, $\text{Corr}(\Delta\text{Spread}, \Delta F)$, computed from annual averages of the simulated quarterly series (Appendix E). The vertical axis reports the empirical correlation between annual changes in EMBI spreads and annual changes in the global factor of Section 2 over 1993–2023. Both axes therefore measure the same object. Annual changes are used because low-frequency movements dominate the observed series in levels over samples of at most 31 years. Correct predictions correspond to observations in the lower-left and upper-right quadrants.

Finally, Table 6 connects the calibrated exposures to the reduced-form evidence. Column (1) interacts the global innovation with the calibrated strength of each channel. The terms-of-trade coefficient is negative and statistically significant: countries with larger model-implied trade-income gains experience larger declines in sovereign spreads. This matches the sign of the net-shock coefficient in Section 2. The price-index coefficient has the predicted positive sign but is imprecisely estimated.

Column (2) combines the two channels in the model-implied consumption exposure, $\hat{\Delta}_i = -\psi_i + \tilde{\nu}_i$, which measures how a country's consumption responds to a positive global shock. The interaction between the global innovation and $\hat{\Delta}_i$ is negative and statistically significant: following a positive innovation, countries with more negative $\hat{\Delta}_i$ experience larger spread increases, while countries with more positive $\hat{\Delta}_i$ experience smaller increases or declines. The model-implied exposure, built from trade and production data without using any spread moment, thus helps explain cross-country differences in the observed response of spreads.

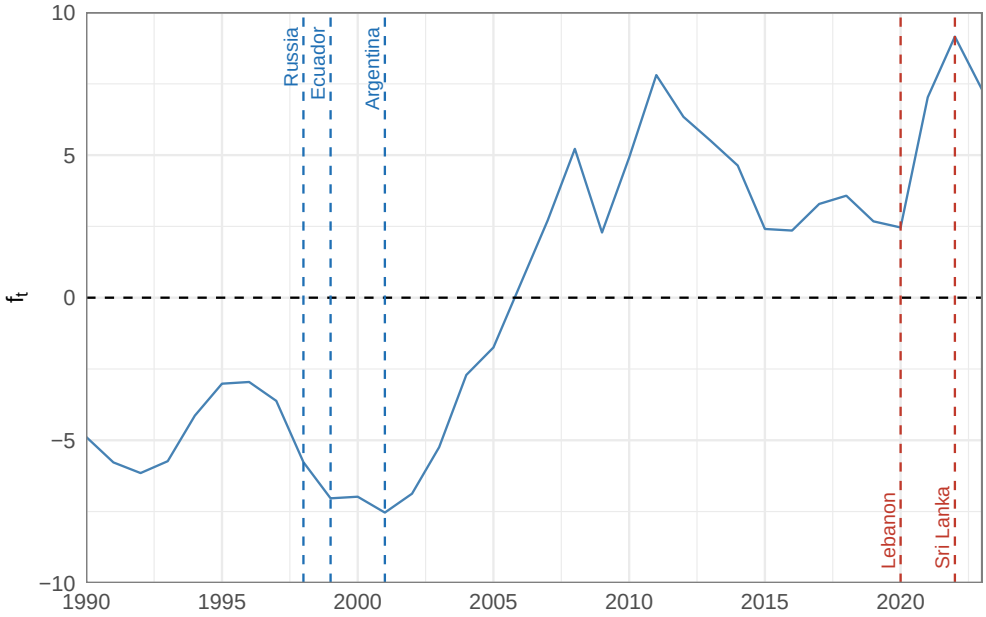
Table 6: Model-Implied Exposure and Sovereign Spreads.

	Δ Spread	
	(1)	(2)
Model Terms-of-Trade Shock	-215.7**	
	(84.3)	
Model Price-Index Shock	924.4	
	(1,143.2)	
Global Shock × Model-Implied Cons. Exposure		-205.6**
		(83.4)
Observations	572	572
R^2	0.30	0.30
Country Fixed Effects	✓	✓
Year Fixed Effects	✓	✓

Note: The model terms-of-trade and price-index shocks combine the global commodity-price innovation with the corresponding country-specific exposure parameters calibrated in the quantitative model. Column (2) interacts the global shock with the signed model-implied combined exposure, $\hat{\Delta}_i = -\psi_i + \tilde{\nu}_i$, which captures both the magnitude and direction of the net effect of the two channels at the reference point (equation (26)). Both specifications include country and year fixed effects and control for real GDP growth, CPI inflation, nominal exchange-rate growth, and the lagged net commodity-trade share. Standard errors are clustered at the country level and reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Taken together, the evidence supports the model's mechanism. Calibrated trade exposure helps explain both the direction and the cross-country variation of spread responses, and does so in particular for exporter-type economies, where the correspondence is closest. The price-index channel is consistent with complementary evidence from Section 2: exposure common to exports and imports is associated with higher spreads and weaker consumption growth. Its coefficient in Table 6 has the predicted sign but is imprecise, as expected given that ψ_i varies little across countries. Differences in trade structure therefore account for most of the model-implied heterogeneity in sovereign-risk transmission, while the price-index channel operates as a common force.

Figure 10: Global Commodity Factor and Sovereign Default Episodes.



Note: The line shows the global commodity factor f_t displayed in Figure 2(a). Vertical lines mark default events, in blue for exporter-type countries and red for importer-type countries.

4.5 Counterfactual Commodity Baskets

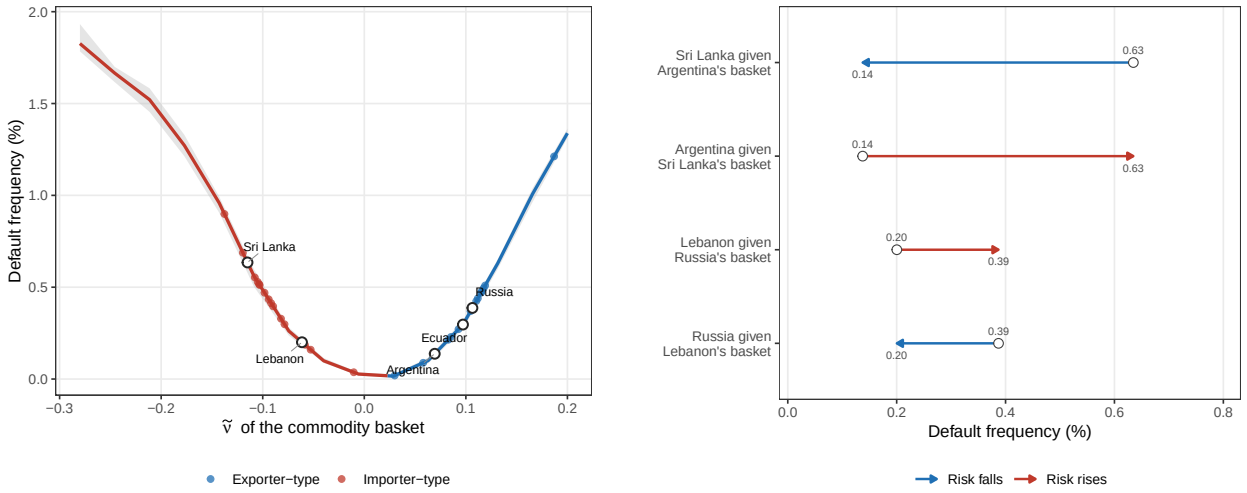
A central implication of the model is that sovereign risk depends not only on the realization of global commodity-price shocks, but also on how a country's commodity basket maps those shocks into real consumption and repayment capacity. We now isolate that mechanism directly.

Because the global factor is common to all countries, we can study the role of the commodity basket by holding fixed the factor process and all other model parameters and varying only the exposure pair $(\psi, \tilde{\nu})$. We solve the model at every point of a two-dimensional grid spanning the empirical ranges of both objects. The grid lets us vary ψ and $\tilde{\nu}$ independently, which isolates the contribution of each. Each of the 37 countries is a point on the surface. A basket reassignment, by contrast, moves a country to another country's pair, since a basket determines both exposures at once. Figure 11 reports the implied default frequency per quarter. Panel (a) plots default frequency against $\tilde{\nu}$, averaged over ψ, and locates each of the 37 countries. Panel (b) reassigns baskets: it compares each country's default frequency under its own basket with the frequency it would have under another country's basket.

Three results stand out. First, default risk varies mainly with the terms-of-trade exposure $\tilde{\nu}$ and little with the price-index exposure ψ. This extends the pattern in Figure 8, where

ψ_i is tightly clustered across countries: even moving ψ over its full empirical range changes default frequency only modestly. In panel (a), the shaded band, which gives the range across ψ, is narrow.

Figure 11: Counterfactual Commodity Baskets.



(a) Where each country sits.

(b) Reassigning a basket.

Note: The figure reports model-implied default frequency, the percentage of quarters in which a default occurs, on a grid spanning the empirical ranges of ψ and $\tilde{\nu}$, holding all other parameters fixed. Panel (a) plots default frequency against $\tilde{\nu}$, averaged across values of ψ; the shaded band gives the range across ψ, and the color indicates exporter- or importer-type exposure. Points correspond to the 37 sample countries. Panel (b) compares each country's default frequency under its own commodity basket (circle) with the counterfactual frequency obtained by assigning it another country's basket (arrowhead).

Second, the level of default risk depends on the magnitude of $\tilde{\nu}$, not on its sign. Default frequency is lowest where $\tilde{\nu}$ is close to zero, that is, where the basket is approximately balanced with respect to the global factor, and it rises as $\tilde{\nu}$ moves away from zero in either direction (panel (a)). In the model, where the factor is the only source of risk, a balanced basket insulates trade income from it, so few movements in global prices push the country toward default. Assigning Sri Lanka's basket to Argentina illustrates this (panel (b)): Argentina's model-implied default frequency rises from 0.14% to 0.63% per quarter, while the reverse reassignment lowers Sri Lanka's from 0.63% to 0.14%. The relevant margin is therefore how directional the basket is, not whether the country is an exporter or an importer: an exporter with a large exposure can be riskier than an importer with a small one. This is a comparative static of a model without hedging or portfolio choice, but it suggests that vulnerability depends on how exposed trade income is to the factor, not on a particular trade orientation.

Third, the sign of $\tilde{\nu}$ matters for a different object: the direction of the response. It determines whether a country is exporter-type or importer-type, and in the model spreads fall with the global factor in the former and rise in the latter. Assigning Russia's basket to Lebanon shows both properties at once. It turns Lebanon from importer-type to exporter-type and reverses the sign of its spread cyclicality, while its default frequency rises from 0.20% to 0.39% per quarter because Russia's terms-of-trade exposure is larger in absolute value. The commodity basket thus shapes sovereign risk along two separate dimensions: the sign of exposure determines when in the global commodity cycle risk is most acute, and its magnitude determines how much risk the basket creates. Assessing an economy's vulnerability therefore requires both.

5 Conclusion

This paper asks why common global commodity-price shocks generate different sovereign-risk responses across countries. We show that the answer depends on how each country's trade and consumption structure transmits the same global price movement into external resources and the cost of consumption. As a result, a global commodity boom can improve repayment capacity in one economy while weakening it in another, even though both are exposed to the same underlying shock. The relevant object is therefore not commodity prices alone, but the interaction between the global commodity cycle and country-specific exposure.

Empirically, we combine a common global commodity-price factor with detailed country trade baskets to construct country-specific exposure. Decomposing the resulting export and import shocks into net and gross components reveals two forces: improvements in net trade exposure lower sovereign spreads and strengthen external resources, while exposure common to both sides of trade is associated with higher spreads and weaker consumption. The model interprets this asymmetry through a terms-of-trade channel, which changes trade income, and a price-index channel, which changes the cost of the consumption basket.

Quantitatively, the model shows that cross-country differences in sovereign risk are driven mainly by terms-of-trade exposure. Its sign determines whether sovereign risk rises or falls with the global commodity factor, while its magnitude governs how much default risk the basket creates. Because price-index exposure varies relatively little across countries, most of the heterogeneity in sovereign-risk dynamics comes from differences in trade exposure. Counterfactual commodity-basket exercises reinforce this distinction between the direction and the strength of vulnerability.

These results have direct implications for sovereign-risk assessment and international financial policy. Commodity booms and busts should not be read as common risk signals across emerging economies. Surveillance and contingent financing should instead account for both the direction and the magnitude of each country's exposure to the global commodity factor. The distinction also matters for commodity-risk management: policies designed only around whether a country is an aggregate commodity exporter or importer can miss substantial heterogeneity within those groups. Finally, the model implies that the terms-of-trade channel is amplified when resources available for consumption are low, pointing to a potentially important role for fiscal, reserve, and liquidity buffers in highly exposed economies. Establishing the empirical importance of this state dependence is a natural direction for future work.

More generally, this paper demonstrates how the sovereign-risk effects of global commodity-price shocks are shaped by country-specific economic structure. Assessing sovereign vulnerability therefore requires accounting for both the direction and the magnitude of each country's exposure to the global commodity cycle, which depend on how its trade and consumption structure transmits that cycle into resources, consumption, and repayment incentives.

References

Adler, G. & Sosa, S. (2011), Commodity price cycles: The perils of mismanaging the boom, IMF Working Paper 2011/283, International Monetary Fund.

Aguiar, M. & Amador, M. (2014), Sovereign debt, *in* G. Gopinath, E. Helpman & K. S. Rogoff, eds, 'Handbook of International Economics', Vol. 4, Elsevier, pp. 647–687.

Aguiar, M. & Gopinath, G. (2006), 'Defaultable debt, interest rates and the current account', *Journal of International Economics* **69**(1), 64–83.

Alberola, E. & Benigno, G. (2017), 'Revisiting the commodity curse: A financial perspective', *Journal of International Economics* **108**, S87–S106.

Alquist, R., Bhattarai, S. & Coibion, O. (2020), 'Commodity-price comovement and global economic activity', *Journal of Monetary Economics* **112**, 41–56.

Arellano, C. (2008), 'Default risk and income fluctuations in emerging economies', *American Economic Review* **98**(3), 690–712.

Baumeister, C. & Hamilton, J. D. (2019), 'Structural interpretation of vector autoregressions with incomplete identification: Revisiting the role of oil supply and demand shocks', *American Economic Review* **109**(5), 1873–1910.

Benguria, F., Saffie, F. & Urzúa, S. (2023), 'The transmission of commodity price supercycles', *Review of Economic Studies* **91**(4), 1923–1955.

Benigno, G. & Fornaro, L. (2014), 'The financial resource curse', *Scandinavian Journal of Economics* **116**(1), 58–86.

Bianchi, J., Ottonello, P. & Presno, I. (2023), 'Fiscal stimulus under sovereign risk', *Journal of Political Economy* **131**(9), 2328–2369.

Boehm, H., Eichler, S. & Giessler, S. (2021), 'What drives the commodity-sovereign risk dependence in emerging market economies?', *Journal of International Money and Finance* **111**, 102308.

Borusyak, K., Hull, P. & Jaravel, X. (2022), 'Quasi-experimental shift-share research designs', *Review of Economic Studies* **89**(1), 181–213.

Byrne, J. P., Fazio, G. & Fiess, N. (2013), 'Primary commodity prices: Co-movements, common factors and fundamentals', *Journal of Development Economics* **101**, 16–26.

Corden, W. M. & Neary, J. P. (1982), 'Booming sector and de-industrialisation in a small open economy', *Economic Journal* **92**(368), 825–848.

Cuadra, G. & Sapriza, H. (2006), Sovereign default, terms of trade and interest rates in emerging markets, Working Paper 2006-01, Banco de México.

Delle Chiaie, S., Ferrara, L. & Giannone, D. (2022), 'Common factors of commodity prices', *Journal of Applied Econometrics* **37**(3), 461–476.

Di Pace, F., Juvenal, L. & Petrella, I. (2025), 'Terms-of-trade shocks are not all alike', *American Economic Journal: Macroeconomics* **17**(2), 24–64.

Drechsel, T. & Tenreyro, S. (2018), 'Commodity booms and busts in emerging economies', *Journal of International Economics* **112**, 200–218.

Eaton, J. & Gersovitz, M. (1981), 'Debt with potential repudiation: Theoretical and empirical analysis', *Review of Economic Studies* **48**(2), 289–309.

Espino, E., Kozlowski, J., Martin, F. M. & Sánchez, J. M. (2025), 'Domestic policies and sovereign default', *American Economic Journal: Macroeconomics* **17**(3), 74–113.

Esquivel, C. (2024), 'Underinvestment and capital misallocation under sovereign risk', *Journal of International Economics* **151**, 103973.

Esquivel, C. (2025), 'Sovereign risk and Dutch disease', *Journal of Monetary Economics* **149**, 103663.

Fernández, A., González, A. & Rodríguez, D. (2018), 'Sharing a ride on the commodities roller coaster: Common factors in business cycles of emerging economies', *Journal of International Economics* **111**, 99–121.

Gelos, R. G., Sahay, R. & Sandleris, G. (2011), 'Sovereign borrowing by developing countries: What determines market access?', *Journal of International Economics* **83**(2), 243–254.

Hamann, F., Mendez-Vizcaino, J. C., Mendoza, E. G. & Restrepo-Echavarria, P. (2023), Natural resources and sovereign risk in emerging economies: A curse and a blessing, NBER Working Paper 31058, National Bureau of Economic Research.

Hilscher, J. & Nosbusch, Y. (2010), 'Determinants of sovereign risk: Macroeconomic fundamentals and the pricing of sovereign debt', *Review of Finance* **14**(2), 235–262.

Känzig, D. R. (2021), 'The macroeconomic effects of oil supply news: Evidence from OPEC announcements', *American Economic Review* **111**(4), 1092–1125.

Kehoe, T. J. & Ruhl, K. J. (2008), 'Are shocks to the terms of trade shocks to productivity?', *Review of Economic Dynamics* **11**, 804–819.

Kilian, L. (2009), 'Not all oil price shocks are alike: Disentangling demand and supply shocks in the crude oil market', *American Economic Review* **99**(3), 1053–1069.

Kilian, L. & Murphy, D. P. (2014), 'The role of inventories and speculative trading in the global market for crude oil', *Journal of Applied Econometrics* **29**(3), 454–478.

Kohn, D., Leibovici, F. & Tretvoll, H. (2021), 'Trade in commodities and business cycle volatility', *American Economic Journal: Macroeconomics* **13**(3), 173–208.

Kose, M. A. (2002), 'Explaining business cycles in small open economies: How much do world prices matter?', *Journal of International Economics* **56**(2), 299–327.

Lopez-Martin, B., Leal, J. & Martinez Fritscher, A. (2019), 'Commodity price risk management and fiscal policy in a sovereign default model', *Journal of International Money and Finance* **96**, 304–323.

Mendoza, E. G. (1995), 'The terms of trade, the real exchange rate, and economic fluctuations', *International Economic Review* **36**(1), 101–137.

Min, H. G. (1998), Determinants of emerging market bond spread: Do economic fundamentals matter?, Policy Research Working Paper 1899, The World Bank.

Miranda-Agrippino, S. & Rey, H. (2020), 'The global financial cycle after Lehman', *AEA Papers and Proceedings* **110**, 523–528.

Müller, K., Xu, C., Lehbib, M. & Chen, Z. (2025), The global macro database: A new international macroeconomic dataset, NBER Working Paper 33714, National Bureau of Economic Research.

Ottonello, P. & Perez, D. J. (2019), 'The currency composition of sovereign debt', *American Economic Journal: Macroeconomics* **11**(3), 174–208.

Pindyck, R. S. & Rotemberg, J. J. (1990), 'The excess co-movement of commodity prices', *Economic Journal* **100**(403), 1173–1189.

Reinhart, C. M., Reinhart, V. & Trebesch, C. (2016), 'Global cycles: Capital flows, commodities, and sovereign defaults, 1815–2015', *American Economic Review* **106**(5), 574–580.

Rey, H. (2015), Dilemma not trilemma: The global financial cycle and monetary policy independence, NBER Working Paper 21162, National Bureau of Economic Research.

Roch, F. (2019), 'The adjustment to commodity price shocks', *Journal of Applied Economics* **22**(1), 437–467.

Schmitt-Grohé, S. & Uribe, M. (2018), 'How important are terms-of-trade shocks?', *International Economic Review* **59**(1), 85–111.

Shousha, S. (2016), Macroeconomic effects of commodity booms and busts: The role of financial frictions, Working paper, Federal Reserve Board.

Sosa-Padilla, C. & Sturzenegger, F. (2023), 'Does it matter how central banks accumulate reserves? Evidence from sovereign spreads', *Journal of International Economics* **140**, 103711.

Tauchen, G. (1986), 'Finite state Markov-chain approximations to univariate and vector autoregressions', *Economics Letters* **20**(2), 177–181.

West, K. D. & Wong, K.-F. (2014), 'A factor model for co-movements of commodity prices', *Journal of International Money and Finance* **42**, 289–309.

Appendix

A Empirical Appendix

A.1 Commodity List

Table 7 lists the 41 internationally traded commodities used to construct the global commodity-price factor, the country-specific exposures, and the calibrated model. Commodity prices are taken from the *IMF Primary Commodity Prices Database*, with the exception of oranges, for which we use the *World Bank Pink Sheet* series.

Table 7: List of Commodities by Category.

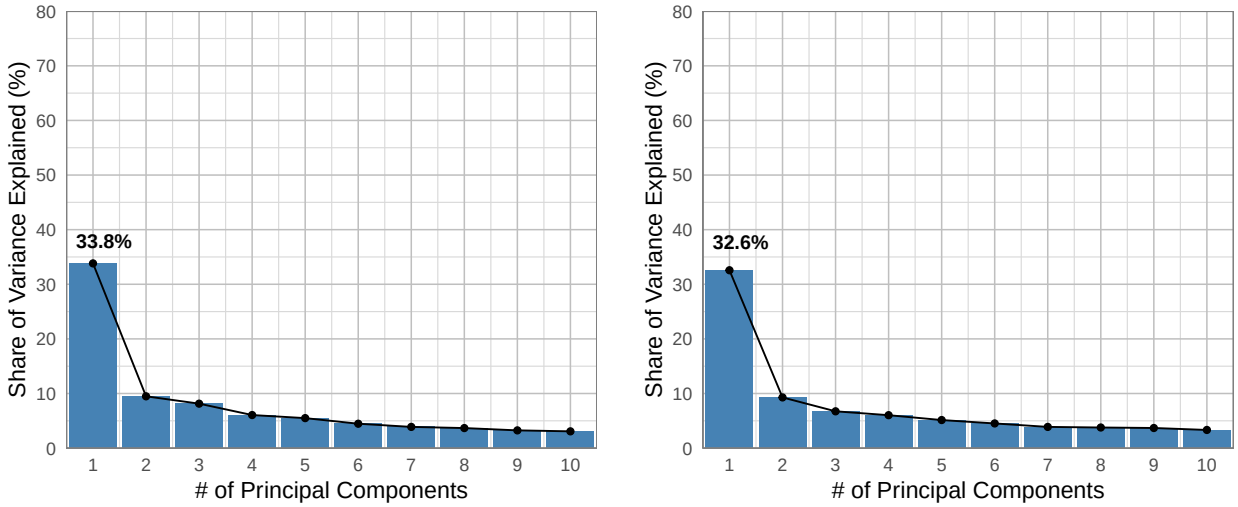
Category	**Commodities**
Energy	Crude oil, natural gas, coal.
Metals & Minerals	Aluminum, copper, iron ores, lead, nickel, tin, uranium, zinc.
Agriculture	Bananas, barley, beef, chicken, cocoa, coffee, corn, fish, groundnuts, lamb, olive oil, oranges, palm oil, pork, rapeseed oil, rice, soybean oil, soybeans, sugar, sunflower seed oil, tea, wheat.
Other Primary	Cotton, hard logs, soft logs, hard sawnwood, soft sawnwood, hides, natural rubber, wool.

A.2 Alternative PCA Specifications

We repeat the PCA on two transformations of commodity prices: annual log price changes and commodity-specific autoregressive (AR) residuals. Both transformations use up one annual observation, so these specifications cover 1991–2023, one year less than the 1990–2023 baseline. Taking annual changes removes the low-frequency, trend-like movements in price levels. AR residuals go further and remove the part of each commodity's price change that is predictable from its own past. Figure 12 shows that the first component explains 33.8% and 32.6% of total variation in the two cases, compared with 66.9% for log price levels over the same 1991–2023 sample (66.6% over 1990–2023 in Figure 1). This drop indicates that much of the comovement in levels reflects persistent global movements, while commodity-specific disturbances account for a larger share of variation at shorter horizons. In both cases, the

first component remains the largest, explaining more than three times as much variation as the second.

Figure 12: Variance Explained by Principal Components under Alternative PCAs.



(a) Annual log price changes

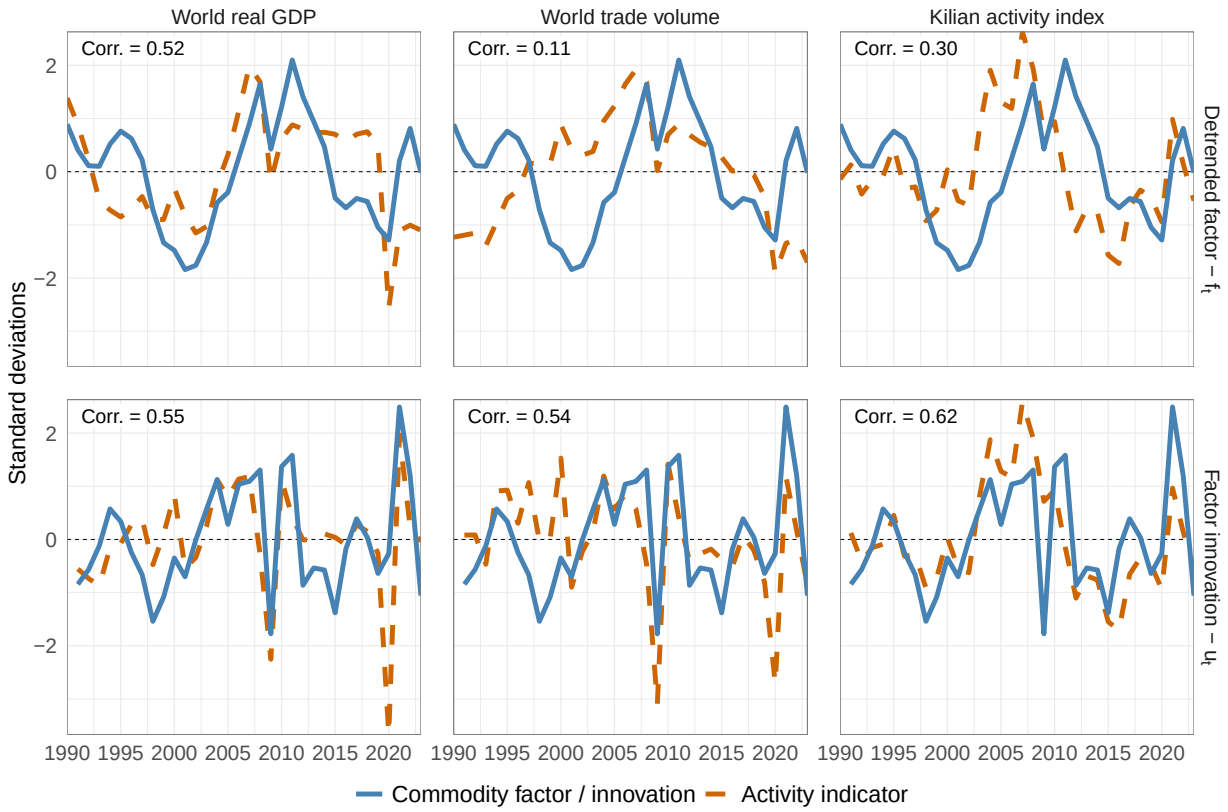
(b) Autoregressive residuals

Note: Panel (a) reports the share of total variation in standardized annual log price changes explained by each of the first ten principal components for the 41 commodities over 1991–2023. Panel (b) reports the corresponding shares for standardized commodity-specific AR residuals. In both panels, each series is demeaned and scaled to unit variance before extracting the principal components, as in Figure 1. The sample starts in 1991 because both transformations use up the first annual observation.

A.3 The Global Commodity-Price Factor and Global Economic Activity

Figure 13 compares the global commodity-price factor and its innovations with three indicators of global economic activity over 1990–2023: world real GDP growth, world trade-volume growth, and the Kilian index of global real economic activity. The factor innovations are positively correlated with all three indicators, with correlations of 0.55, 0.54, and 0.62, respectively. Correlations with the detrended factor remain positive, though smaller, ranging from 0.11 for world trade to 0.52 for world GDP. These patterns indicate that the global commodity-price factor captures broad movements in global economic activity, with the factor innovations closely aligned with fluctuations in global demand.

Figure 13: Global Commodity Prices and Global Economic Activity.



Note: The upper panels plot the linearly detrended global commodity-price factor f_t against detrended log indices of world real GDP and world trade volume, and against the Kilian activity index, over 1990–2023. The log indices are constructed by cumulating $\log(1 + g_t/100)$, where g_t is the corresponding IMF annual growth rate. The lower panels plot the innovations $u_t = f_t - \widehat{\rho}_f f_{t-1}$, obtained from the AR(1) specification in equation (2), against annual growth in world real GDP and world trade volume, and against the Kilian index. The Kilian index is averaged across months within each year. All series are standardized. Reported correlations are contemporaneous Pearson correlations based on 33 annual observations. Sources: IMF World Economic Outlook and the Federal Reserve Bank of Dallas, accessed through FRED.

B Derivations of the Aggregate Environment

This appendix derives the aggregate consumption equation used in Section 3.1 and clarifies the role of non-homothetic preferences. We first derive the baseline environment and then examine its Cobb–Douglas limit. Section S2 of the Supplemental Material discusses the measurement of net endowments and an alternative CES specification.

B.1 Baseline Aggregate Environment

The Stone–Geary aggregator introduced in Section 3.1 is:

$$c_t = \prod_{j\in[0,1]} [c_t(j) - \bar{c}(j)]^{\lambda(j)}, \qquad \int_0^1 \lambda(j)\, dj = 1,$$

where $\lambda(j) \geq 0$ is the weight assigned to good j. Taking logs yields:

$$\ln c_t = \int_0^1 \lambda(j) \ln[c_t(j) - \bar{c}(j)]\, dj.$$

Expenditure minimization yields the individual demand functions:

$$c_t(j) = \bar{c}(j) + \frac{\lambda(j)}{P_t(j)}\Big(E_t - \textstyle\int_0^1 P_t(j')\bar{c}(j')\, dj'\Big),$$

where E_t denotes aggregate expenditure. Substituting these demands into the consumption aggregator gives:

$$c_t = \Gamma^c (P_t^C)^{-1} \left(E_t - \int_0^1 P_t(j)\bar{c}(j)\, dj\right), \tag{27}$$

where

$$P_t^C = \exp\left(\int_0^1 \lambda(j) \ln P_t(j)\, dj\right), \qquad \Gamma^c = \exp\left(\int_0^1 \lambda(j) \ln \lambda(j)\, dj\right).$$

Using the aggregate resource constraint:

$$E_t = \int_0^1 P_t(j) e(j)\, dj + D_t$$

and defining the net endowment $\tilde{e}(j) \equiv e(j) - \bar{c}(j)$, equation (27) becomes:

$$c_t = \Gamma^c (P_t^C)^{-1} \left(D_t + \int_0^1 P_t(j)\tilde{e}(j)\, dj\right). \tag{28}$$

The factor structure of international commodity prices in equation (1) implies:

$$\ln P_t^C = \mu^C + \boldsymbol{\psi}^\top \mathbf{f}_t + \varepsilon_t^C,$$

where μ^C and $\boldsymbol{\psi}$ are defined in equation (10) and $\varepsilon_t^C = \int_0^1 \lambda(j)\varepsilon_t(j)\, dj$.

Similarly, a first-order approximation of trade income around $f_t = 0$ yields:

$$\int_0^1 P_t(j)\tilde{e}(j)\, dj \approx \bar{Z} + \nu^\top f_t + \zeta_t,$$

where

$$\bar{Z} = \int_0^1 \bar{P}(j)\tilde{e}(j)\,dj, \qquad \nu = \int_0^1 \bar{P}(j)\tilde{e}(j)a(j)\,dj,$$

and $\zeta_t \equiv \int_0^1 \bar{P}(j)\tilde{e}(j)\varepsilon_t(j)\,dj$ collects the good-specific price components.

Setting the commodity-specific price components to zero gives:

$$c_t = \Gamma^c e^{-\mu^C - \psi^\top f_t}\left(D_t + \bar{Z} + \nu^\top f_t\right). \tag{29}$$

This expression corresponds to equation (12) in the main text. It is exact given the first-order approximation of trade income in equation (11): the aggregation under Stone–Geary preferences, including the price index, introduces no further approximation.

B.2 Stone–Geary and the Cobb–Douglas Limit

To clarify the role of the non-homotheticity introduced by subsistence, we consider the Cobb–Douglas limit obtained by setting $\bar{c}(j) \equiv 0$ for every good. The net endowment then coincides with the physical endowment, $\tilde{e}(j) = e(j) \geq 0$, while expenditure shares become constant and equal to $\lambda(j)$.

Let $Y = \int_0^1 \bar{P}(j)e(j)\,dj$ denote endowment income at reference prices, and define the endowment-weighted factor exposure as $\bar{a}_k^e = \int_0^1 [\bar{P}(j)e(j)/Y]a_k(j)\,dj$, with $\bar{a}_k^c \equiv \psi_k$ denoting the corresponding consumption-weighted exposure.

Exposures in the limit. Subsistence enters the aggregate environment only through the net endowment $\tilde{e}(j) = e(j) - \bar{c}(j)$, so the limit changes $\bar{Z}$ and ν and leaves ψ untouched. At the reference state, the consumption exposure in Proposition 1 becomes:

$$\Delta_k = -\bar{a}_k^c + \frac{Y}{\bar{D} + Y}\bar{a}_k^e, \qquad \Delta_k\Big|_{\bar{D}=0} = \bar{a}_k^e - \bar{a}_k^c, \tag{30}$$

where $\bar{D} \equiv D(\mathbf{0}, b)$ denotes the resource flow at the reference state. Consumption can therefore fall under Cobb–Douglas preferences if the price-index effect exceeds the positive income effect.

The distinction from Stone–Geary concerns the sign and magnitude of terms-of-trade exposure. Under Cobb–Douglas, $\nu_k = \int_0^1 \bar{P}(j)e(j)a_k(j)\,dj$. If all factor loadings are non-negative, ν_k cannot be negative. In our application, 39 of the 41 first-component loadings are

positive and the two negative loadings are small[27]; none of the calibrated commodity baskets generates negative ν_k in the Cobb–Douglas limit. By contrast, Stone–Geary preferences allow $\tilde{e}(j) < 0$ when subsistence exceeds the endowment of a good. Consequently, ν_k can be negative, so rising global commodity prices may simultaneously reduce trade income and increase consumption costs. Cobb–Douglas also restricts the magnitude of normalized exposure. Since endowment weights are non-negative and sum to one, ν_k/Y lies within the range of factor loadings. Under Stone–Geary, signed net endowments and cancellation in aggregate trade imply that normalized exposure $\tilde{\nu}_k$ (equation (25)) need not satisfy this bound. In particular, normalized exposure can be very large when $|\bar{Z}|$ is small. In the calibration, the loadings span $[-0.011, 0.183]$ and every ψ_i lies within this interval, whereas Mexico ($\tilde{\nu}_i = 0.19$) lies outside it, a value that a Cobb–Douglas economy with non-negative endowment weights cannot generate.

Role of subsistence. Subsistence requirements affect both the sign and the strength of the terms-of-trade channel. Holding the endowment fixed, a larger $\bar{c}(j)$ reduces the net endowment and hence lowers $\bar{Z}$. At the reference state, this raises the amplification term $\kappa_c = 1/(\bar{D} + \bar{Z})$, so a given change in trade income has a larger effect on consumption when the resources available above subsistence are smaller. The price-index exposure ψ is unaffected because it depends only on the composition of the consumption basket. It is useful to distinguish the objects taken as exogenous from those determined in equilibrium. The commodity-specific endowment and subsistence requirements, $e(j)$ and $\bar{c}(j)$, determine the net endowment profile $\tilde{e}(j)$ and therefore the exposure objects $\bar{Z}$ and ν. The model does not seek to explain why a country is a net buyer or seller of a particular commodity. By contrast, aggregate resources available above subsistence, $D_t + \bar{Z} + \boldsymbol{\nu}^\top \mathbf{f}_t$, are endogenous because the resource flow D_t depends on debt policy and the price of credit. Default affects these resources by eliminating market access and, through $\chi < 1$, reducing consumption supported by the remaining trade income. These equilibrium adjustments determine how the calibrated commodity exposures translate into repayment incentives.

[27] The exceptions are hides (-0.011) and lamb (-0.0005), compared with a maximum loading of 0.183. Obtaining $\nu_k < 0$ under Cobb–Douglas would require more than 85% of endowment value to be concentrated in hides alone, with the remainder distributed at the average loading.

C Proofs and Extensions

C.1 Proof of Proposition 1

Proof. Let $R(\mathbf{f}, b) = D(\mathbf{f}, b) + \bar{Z} + \boldsymbol{\nu}^\top \mathbf{f} > 0$. Taking logs of aggregate consumption gives:

$$\ln c(\mathbf{f}, b) = \ln \Gamma^c - \mu^C - \boldsymbol{\psi}^\top \mathbf{f} + \ln R(\mathbf{f}, b).$$

Holding the resource flow D fixed:

$$\Delta_k(\mathbf{f}, b) = \left. \frac{\partial \ln c(\mathbf{f}, b)}{\partial f_k} \right|_D = -\psi_k + \frac{\nu_k}{R(\mathbf{f}, b)}.$$

This establishes the price-index and terms-of-trade decomposition in part (i).

At the reference state, $R(\mathbf{0}, b) = D(\mathbf{0}, b) + \bar{Z} > 0$. For $\psi_k > 0$, consumption increases with f_k if and only if $\nu_k / R(\mathbf{0}, b) > \psi_k$, which requires $\nu_k > 0$. This proves part (ii).

Finally, the magnitude of the terms-of-trade channel, $|\nu_k| / R(\mathbf{0}, b)$, increases as available resources decline. For $\nu_k < 0$ and $\psi_k > 0$, consumption exposure is negative at every feasible resource level. For $\nu_k > 0$, its sign changes at:

$$D(\mathbf{0}, b) = D_k^* \equiv \frac{\nu_k}{\psi_k} - \bar{Z},$$

provided this threshold lies in the feasible range, with $\Delta_k(\mathbf{0}, b) > 0$ if and only if $D(\mathbf{0}, b) < D_k^*$. This establishes part (iii).

The proposition concerns the direct consumption response holding D fixed. The full equilibrium derivative additionally contains $[\partial D(\mathbf{f}, b)/\partial f_k]/R(\mathbf{f}, b)$. With b' fixed, this term is, by the pricing condition (15),

$$\left. \frac{\partial D(\mathbf{f}, b)}{\partial f_k} \right|_{b'} = -\frac{\partial q(\mathbf{f}, b')}{\partial f_k} b' = \frac{b'}{1 + r^*} \frac{\partial \delta(\mathbf{f}, b')}{\partial f_k}.$$

It is zero when the default set $\mathcal{D}(b')$ is empty, in particular when $b' \geq 0$, since default is then impossible. Otherwise a change in f_k shifts the conditional distribution of next period's factors, so δ moves with f_k even though $\mathcal{D}(b')$ does not; the term is largest where δ responds most sharply to the factor and the country holds large debt, that is, near the default frontier. The envelope theorem removes the effect of the induced change in b' on the value function, but not on consumption: the full derivative of D also contains $-q\, \partial b'/\partial f_k$, which is generally non-zero. We therefore read $\Delta_k(\mathbf{f}, b)$ as an impact elasticity, and rely on Corollary 1 and

Lemma 1 for the behavior of the full derivative. In every calibration the default frontier is monotone in the factor (Figure 4 and Figures S2–S5 of the Supplemental Material), and the sign of its slope is given by the type. □

C.2 Global Factors, Default Sets, and Spreads

Proposition 1 characterizes the direct effect of global commodity-price factors on consumption. Translating this effect into sovereign spreads requires additional conditions on relative repayment incentives and the transition of global factors.

Corollary 1 (Global factors, default sets, and spreads)**.** *Let* $G(\mathbf{f}, b) \equiv V^r(\mathbf{f}, b) - V^d(\mathbf{f})$ *denote the repayment–default value difference. Suppose that:*

(i) Repayment satisfies a single-crossing rule, with a cutoff $b^*(\mathbf{f})$ *such that repayment occurs if and only if* $b \geq b^*(\mathbf{f})$*, and* G *is strictly increasing in* b *at the cutoff.*

(ii) At the relevant default frontier, the response of relative repayment incentives has the same sign as consumption exposure:

$$\operatorname{sgn}\left(\frac{\partial G(\mathbf{f}, b)}{\partial f_k}\right) = \operatorname{sgn}\left(\Delta_k(\mathbf{f}, b)\right).$$

(iii) The factor transition is increasing in f_k *in the sense of first-order stochastic dominance (in the one-factor case,* $\rho_f > 0$*).*

If $\Delta_k > 0$ *throughout the relevant default frontier, a marginal increase in* f_k *weakly lowers the repayment cutoff, reduces the one-period default probability, raises bond prices, and lowers sovereign spreads. The opposite comparative statics hold when* $\Delta_k < 0$ *throughout the relevant frontier.*

Proof. At the repayment cutoff, $G(\mathbf{f}, b^*(\mathbf{f})) = 0$. Implicit differentiation gives:

$$\frac{\partial b^*(\mathbf{f})}{\partial f_k} = -\frac{\partial G/\partial f_k}{\partial G/\partial b}.$$

Since $\partial G/\partial b > 0$, assumption (ii) implies that the cutoff decreases when $\Delta_k > 0$. The default region therefore contracts as f_k rises.

Under assumption (iii), this contraction implies that the conditional probability of default weakly decreases. The lenders' zero-profit condition,

$$q(\mathbf{f}, b') = \frac{1 - \delta(\mathbf{f}, b')}{1 + r^*},$$

then implies higher bond prices and lower sovereign spreads. The reverse argument applies when consumption exposure is negative, since the default region then expands with f_k and assumption (iii) raises the probability of entering it. □

C.3 Global Comparative Statics Under Bounded Resources

Proposition 1 characterizes consumption exposure locally. The following result gives sufficient conditions under which the sign of the direct response is uniform across states.

Lemma 1 (Global comparative statics under bounded resources)**.** *Let $\mathcal{S}$ be a set of admissible states such that resources available for consumption satisfy:*

$$0 < R_{\min} \leq R(\mathbf{f}, b) \equiv D(\mathbf{f}, b) + \bar{Z} + \boldsymbol{\nu}^\top \mathbf{f} \leq R_{\max}$$

for all $(\mathbf{f}, b) \in \mathcal{S}$. Suppose $\psi_k > 0$. Then:

(i) Terms-of-trade dominance. If $\nu_k > 0$ and

$$\frac{\nu_k}{R_{\max}} > \psi_k,$$

consumption exposure is positive throughout $\mathcal{S}$.

(ii) Negative terms-of-trade exposure. If $\nu_k < 0$, consumption exposure is negative throughout $\mathcal{S}$.

Proof. Fix $(\mathbf{f}, b) \in \mathcal{S}$. By the proof of Proposition 1, holding the resource flow fixed, consumption exposure is:

$$\left.\frac{\partial \ln c(\mathbf{f}, b)}{\partial f_k}\right|_D = -\psi_k + \frac{\nu_k}{R(\mathbf{f}, b)}.$$

Since $R(\mathbf{f}, b) \in [R_{\min}, R_{\max}]$ with $R_{\min} > 0$, the second term is bounded and has the sign of ν_k.

If $\nu_k > 0$, the upper bound on resources implies

$$\Delta_k(\mathbf{f}, b) \geq -\psi_k + \frac{\nu_k}{R_{\max}} > 0.$$

If $\nu_k < 0$, both terms are negative, so $\Delta_k(\mathbf{f}, b) < 0$ throughout $\mathcal{S}$. The corresponding implications for default probabilities and bond prices follow under the conditions of Corollary 1. □

D Cross-Country Calibration of (ψ_i, ν_i)

This appendix details the construction of the country-specific exposure parameters (ψ_i, ν_i) used in the quantitative analysis. The objective is to map the model's net endowments, baseline prices, and consumption weights into observable counterparts for each of the 37 countries and the $J = 41$ commodities in our sample.

The procedure mirrors the structure of the model. We first describe the data and commodity coverage used to implement the construction in Section 4.1, and then report the resulting cross-country estimates and classifications.

D.1 Data Sources and Commodity Coverage

The calibration combines international commodity prices, trade flows, and production quantities for 41 commodities spanning energy, metals, agriculture, forestry, and livestock. Annual international prices are drawn from the IMF Primary Commodity Prices Database, except for oranges, for which we use the World Bank Pink Sheet. Annual export and import values are obtained from UN Comtrade at the SITC Revision 1 level. Production quantities are obtained from FAOSTAT for agriculture and forestry, the U.S. EIA for energy, and USGS and UN Data for metals and minerals.

The detailed crosswalk between SITC codes, FAO items, and production sources is reported in Table S3 of the Supplemental Material. Physical units are harmonized with those of the corresponding international price series. Trade and production data cover the calibration period 1991–2019, while factor loadings are estimated using the full 1990–2023 price sample described in Section 2.

D.2 Construction of Net Exports and Apparent Consumption

This subsection details the construction of the fixed-price net exports and apparent consumption measures used to calibrate country-specific exposures in Section 4.1. We combine the price, trade, and production data described above, valuing all commodity flows at common reference prices over 1991–2019.

Let P_{jt} denote the annual international price of commodity j. Its reference price is the geometric mean over the calibration period:

$$\bar{P}(j) = \exp\left(\frac{1}{29}\sum_{t=1991}^{2019}\log P_{jt}\right).$$

For country i, let Q_{ijt}^{P} denote production quantities and let X_{ijt} and M_{ijt} denote export and import values in current US dollars. We express these flows at fixed reference prices:

$$V_{ijt}^{P,0} = Q_{ijt}^{P}\bar{P}(j), \qquad X_{ijt}^{0} = X_{ijt}\frac{\bar{P}(j)}{P_{jt}}, \qquad M_{ijt}^{0} = M_{ijt}\frac{\bar{P}(j)}{P_{jt}}.$$

Production quantities are converted to units consistent with the international price series. For trade flows, the price adjustment uses each international commodity price as a proxy for prices within the corresponding traded commodity group.

We construct average net exports and apparent consumption as

$$NX_{ij} = \overline{X_{ijt}^{0} - M_{ijt}^{0}}, \qquad C_{ij} = \overline{\max\left\{V_{ijt}^{P,0} - X_{ijt}^{0} + M_{ijt}^{0},\, 0\right\}},$$

where the overbar denotes an average over available annual observations within 1991–2019. Negative annual values of apparent consumption are set to zero before averaging. The resulting measures enter the construction of country-specific exposures in equation (24).

D.3 Country-Level Estimates

Table 8 reports the estimated parameters $(\psi_i, \tilde{\nu}_i)$ and the model-implied consumption exposure defined in equation (26). The estimates display substantial heterogeneity. While ψ_i lies in a narrow range (approximately 0.053 to 0.105), $\tilde{\nu}_i$ varies widely, reflecting differences in trade specialization. As a result, the sign of $\hat{\Delta}_i$ is primarily driven by trade structure. These calibrated parameters underpin the cross-country comparative statics in the main text. The sign of $\tilde{\nu}_i$ determines a country's exporter- or importer-type classification, while the sign of $\hat{\Delta}_i$ indicates whether a positive global commodity-price shock raises or lowers consumption at the reference point.

Table 8: Country-Level Exposure Parameters.

Country	ψ_i	$\tilde{\nu}_i$	$\hat{\Delta}_i$	Type
Argentina	0.072	0.069	−0.003	Exporter
Belize	0.058	0.058	0.000	Exporter
Brazil	0.067	0.083	0.015	Exporter
Bulgaria	0.088	−0.108	−0.196	Importer
Chile	0.064	0.107	0.043	Exporter
China	0.080	−0.116	−0.196	Importer
Colombia	0.080	0.110	0.030	Exporter
Dominican Republic	0.069	−0.082	−0.151	Importer
Ecuador	0.082	0.097	0.015	Exporter
Egypt	0.082	−0.058	−0.140	Importer
El Salvador	0.062	−0.092	−0.154	Importer
Gabon	0.085	0.112	0.026	Exporter
Ghana	0.065	0.085	0.020	Exporter
Hungary	0.083	−0.120	−0.202	Importer
Indonesia	0.071	0.105	0.034	Exporter
Iraq	0.105	0.117	0.012	Exporter
Jamaica	0.072	−0.090	−0.162	Importer
Kazakhstan	0.097	0.115	0.018	Exporter
Lebanon	0.057	−0.061	−0.119	Importer
Malaysia	0.070	0.093	0.023	Exporter
Mexico	0.085	0.187	0.101	Exporter
Pakistan	0.054	−0.079	−0.132	Importer
Panama	0.083	−0.104	−0.186	Importer
Peru	0.073	0.116	0.042	Exporter
Philippines	0.070	−0.098	−0.168	Importer
Poland	0.078	−0.103	−0.181	Importer

(continued)

Country	ψ_i	$\tilde{\nu}_i$	$\hat{\Delta}_i$	Type
Russia	0.088	0.106	0.019	Exporter
Serbia	0.083	−0.114	−0.197	Importer
South Africa	0.087	−0.010	−0.097	Importer
South Korea	0.099	−0.105	−0.204	Importer
Sri Lanka	0.053	−0.115	−0.168	Importer
Tunisia	0.065	−0.053	−0.118	Importer
Turkey	0.060	−0.094	−0.154	Importer
Ukraine	0.084	−0.138	−0.222	Importer
Uruguay	0.060	0.030	−0.030	Exporter
Venezuela	0.100	0.119	0.019	Exporter
Viet Nam	0.058	0.106	0.048	Exporter

Note: ψ_i is the consumption-expenditure-weighted average of factor loadings $a(j)$. The normalized terms-of-trade exposure is $\tilde{\nu}_i = \nu_i / \max\{|\bar{Z}_i|, R_i\}$, where $\nu_i = \sum_j NX_{ij} a(j)$, $\bar{Z}_i = \sum_j NX_{ij}$, and $R_i = 0.1 \sum_j C_{ij}$ (equation (25)). Factor loadings are estimated by PCA using the IMF commodity price panel. The reference period is 1991–2019. Consumption exposure is evaluated at $f = 0$ and $D = 0$, where $\kappa_c = 1$, yielding $\hat{\Delta}_i = \tilde{\nu}_i - \psi_i$. Countries are classified as exporter-type or importer-type according to the sign of $\tilde{\nu}_i$ (equation (25)), which determines the sign of the spread–factor covariance in our simulations: negative for exporter-type economies and positive for importer-type economies. The signs of $\hat{\Delta}_i$ and $\tilde{\nu}_i$ coincide in 35 of the 37 countries. The exceptions are Argentina and Uruguay, where $\tilde{\nu}_i > 0$ but $\hat{\Delta}_i < 0$. In these economies, the terms-of-trade channel can nevertheless dominate in repayment states where $\kappa_c > \psi_i / \tilde{\nu}_i$; together with the income-dependent utility cost of default, this explains their negative spread–factor covariance (Section 4.2 and Appendix C.1). Belize's $\hat{\Delta}_i$ is positive before rounding (0.0002).

E Numerical Implementation

Table 9 reports the discretization, convergence, and simulation settings used across countries. For calibrations with large $|\tilde{\nu}_i|$, value-function iteration may fail to converge as trade income, $\bar{Z} + \nu f$, approaches zero at the outer nodes of the factor grid. In these cases, the solver repeats

the solution on a narrower grid spanning three rather than four unconditional standard deviations on either side of the mean, keeping trade income strictly positive at every node. All baseline country calibrations converge on the four-standard-deviation grid. Bond-price iteration is capped at 500 rounds, a limit that no baseline country calibration reaches. The few points in the counterfactual grid of Section 4.5 that reach this limit are re-solved on the three-standard-deviation grid.

Table 9: Numerical Settings for the Solution and Simulation Algorithm.

Setting	**Symbol**	**Value**
State-Space Discretization		
Debt grid points	n_B	201
Debt grid bounds	$[\underline{b}, \bar{b}]$	$[-0.9,\ 0.9]$
Factor grid points (Tauchen)	n_F	60
Tauchen half-width (unconditional std. dev.)	m_{TH}	4
Tauchen grid center	μ_{TH}	0
Number of global factors	K	1
Convergence Criteria		
Value-function tolerance (sup norm)	tol_V	10^{-5}
Maximum value-function iterations	maxit_V	1,200
Bond-price tolerance (sup norm)	tol_q	10^{-5}
Maximum bond-price iterations	maxit_q	500
Simulation		
Independent histories		500
Periods per history	T	10,000
Event-window half-length		180

Note: The model is solved by nested iteration: an outer loop on the bond-price schedule $q(f, b')$ and an inner value-function iteration, each terminated on the sup-norm criteria above. The factor grid is constructed with the Tauchen (1986) method applied to the estimated AR(1) process for f_t. Event windows are used to construct the dynamics around default episodes reported in Figure 6; retained episodes are separated by at least the window length.

The per-period spread is $s_t = 1/q(f_t, b_{t+1}) - (1 + r^*)$, which we annualize as $(1 + s_t)^4 - 1$. Before computing spread moments, we cap each country's simulated spread at its 99th percentile. Under the annualization formula, annualized spreads rise sharply as bond prices approach zero, allowing a handful of observations near the default frontier to dominate the mean and variance. The cap limits the influence of these extreme observations. Observed

spreads are also subject to selection around default episodes, as sovereigns leave the EMBI during default and re-enter after restructuring. We apply the same capping rule to every country; it affects only reported moments, leaving policy functions, default sets, and bond prices unchanged. For the comparison with EMBI data in Figure 9, we average the simulated quarterly spread and global factor within each year, drop years with a quarter in default or exclusion, and correlate annual changes within each simulated history, averaging across histories.

F Cross-Country Quantitative Results

This appendix section extends the quantitative analysis of Section 4.2 to the full sample of 37 countries. For each country, we solve the model using its country-specific exposure parameters $(\psi_i, \tilde{\nu}_i)$ and simulate 500 histories of 10,000 periods. Countries are classified by the sign of $\tilde{\nu}_i$ (equation (25)), which determines the sign of the simulated spread–factor covariance in every country: negative for exporter-type economies and positive for importer-type economies.

The results are presented in two parts: simulated moments (Table 10) and dynamics around default events (Figure 14). Both present exporter-type countries first, followed by importer-type countries. Within each group, the figure orders countries by decreasing absolute terms-of-trade exposure, $|\tilde{\nu}_i|$, while the table orders them by $\hat{\Delta}_i$.

Default sets (Section S5.1, Figures S2–S5) and bond-price schedules (Section S5.2, Figures S6–S9) for all 37 countries are reported in the separately available Supplemental Material.

F.1 Simulation Moments: Full Sample

Table 10 extends Table 5 to all 37 countries, grouped by exposure type. It retains the same column structure, reporting the mean, standard deviation, and covariance with the global factor for annual spreads (%), debt, and consumption. Moments of the global factor are reported in the table note.

Table 10: Simulation Moments Across All 37 Sample Countries.

	Annual Spread (%)			Debt			Consumption		
Country	Mean	Std	Cov w/ F	Mean	Std	Cov w/ F	Mean	Std	Cov w/ F
Exporter-type (18 countries)									
Mexico	20.45	37.27	-24.48	0.26	0.22	0.20	0.98	0.12	0.12
Viet Nam	2.49	5.25	-2.76	0.38	0.20	0.19	0.99	0.07	0.06
Chile	2.48	5.24	-2.76	0.38	0.20	0.19	0.99	0.06	0.05
Peru	3.03	6.23	-3.50	0.36	0.20	0.19	0.99	0.06	0.05
Indonesia	1.67	2.71	-1.62	0.38	0.20	0.18	0.99	0.05	0.04
Colombia	2.59	5.36	-2.89	0.37	0.20	0.19	0.99	0.05	0.04
Gabon	2.62	5.41	-2.95	0.37	0.20	0.19	0.99	0.05	0.03
Malaysia	1.25	1.95	-1.13	0.42	0.19	0.17	0.99	0.04	0.03
Ghana	1.06	1.66	-0.92	0.44	0.18	0.17	0.99	0.04	0.03
Venezuela	3.07	6.29	-3.57	0.36	0.20	0.19	0.99	0.05	0.03
Russia	1.74	2.73	-1.66	0.39	0.20	0.18	0.99	0.04	0.03
Kazakhstan	2.91	6.14	-3.43	0.37	0.20	0.19	0.99	0.05	0.03
Ecuador	1.38	2.17	-1.25	0.41	0.19	0.18	0.99	0.04	0.02
Brazil	0.95	1.45	-0.76	0.46	0.18	0.16	0.99	0.04	0.02
Iraq	2.90	6.14	-3.38	0.37	0.20	0.19	0.99	0.04	0.02
Belize	0.40	0.45	-0.21	0.57	0.13	0.12	0.99	0.03	0.00
Argentina	0.56	0.66	-0.32	0.52	0.15	0.14	0.99	0.03	0.00
Uruguay	0.10	0.09	-0.02	0.73	0.06	0.05	0.99	0.03	-0.03
Importer-type (19 countries)									
South Africa	0.15	0.14	0.04	0.68	0.08	-0.07	1.00	0.10	-0.10
Tunisia	0.57	0.68	0.34	0.49	0.16	-0.14	1.00	0.13	-0.12
Lebanon	0.72	0.96	0.50	0.46	0.17	-0.15	1.00	0.13	-0.12
Pakistan	1.28	1.94	1.14	0.40	0.19	-0.17	1.00	0.14	-0.14
Egypt	0.72	0.91	0.48	0.46	0.17	-0.16	1.00	0.15	-0.15

(continued)

Country	Mean	Std	Cov w/ F	Mean	Std	Cov w/ F	Mean	Std	Cov w/ F
Dominican Republic	1.43	2.26	1.33	0.38	0.19	-0.18	1.00	0.16	-0.16
El Salvador	2.40	4.95	2.65	0.36	0.19	-0.18	1.00	0.16	-0.16
Turkey	2.42	5.00	2.76	0.36	0.20	-0.18	1.00	0.17	-0.16
Jamaica	1.76	2.69	1.70	0.36	0.19	-0.18	1.00	0.17	-0.17
Sri Lanka	4.14	7.98	4.91	0.31	0.21	-0.19	1.00	0.18	-0.18
Philippines	2.83	5.79	3.23	0.34	0.20	-0.19	1.00	0.18	-0.18
Poland	3.44	7.02	4.08	0.33	0.20	-0.19	1.01	0.19	-0.19
Panama	3.35	6.77	3.95	0.33	0.20	-0.19	1.01	0.20	-0.20
Bulgaria	3.84	7.53	4.59	0.31	0.20	-0.19	1.01	0.21	-0.21
China	4.54	8.65	5.41	0.30	0.21	-0.19	1.01	0.21	-0.21
Serbia	4.39	8.27	5.19	0.30	0.21	-0.19	1.01	0.21	-0.21
Hungary	4.91	9.01	5.78	0.29	0.21	-0.19	1.01	0.22	-0.21
South Korea	3.66	7.12	4.33	0.31	0.20	-0.19	1.01	0.22	-0.22
Ukraine	13.50	29.55	17.56	0.27	0.21	-0.19	1.01	0.24	-0.24

Note: The global factor itself has Mean ≈ 0.00 and Std ≈ 1.01 across all 37 countries.

Cross-country patterns. The full-sample simulations display four regularities.

(i) Sharp separation in spread cyclicality. The spread–factor covariance is negative in every exporter-type country and positive in every importer-type country, ranging from -24.48 in Mexico to 17.56 in Ukraine. Because the type is fixed by the sign of $\tilde{\nu}_i$ before the model is solved, this separation is a quantitative result rather than a feature imposed by the classification (Section 4.1).

(ii) Greater consumption risk in importer-type economies. The standard deviation of consumption averages 0.18 in importer-type countries, compared with 0.05 in exporter-type countries. Its covariance with the global factor is positive in most exporter-type economies, reaching 0.12 in Mexico, slightly negative in Uruguay, and strongly negative in importer-type economies. These patterns reflect the interaction of the two channels: the price-index and

terms-of-trade effects reinforce each other in importer-type countries and partially offset in exporter-type countries.

(iii) High spread variability. Across countries, the standard deviation of spreads ranges from 0.9 to 2.2 times the mean, with a typical ratio of about 1.9. Among importer-type economies, Ukraine has both the highest mean spread (13.50%) and the largest standard deviation (29.55%), followed by Hungary and China. Among exporter-type economies, Mexico has the highest mean spread and the largest positive terms-of-trade exposure.

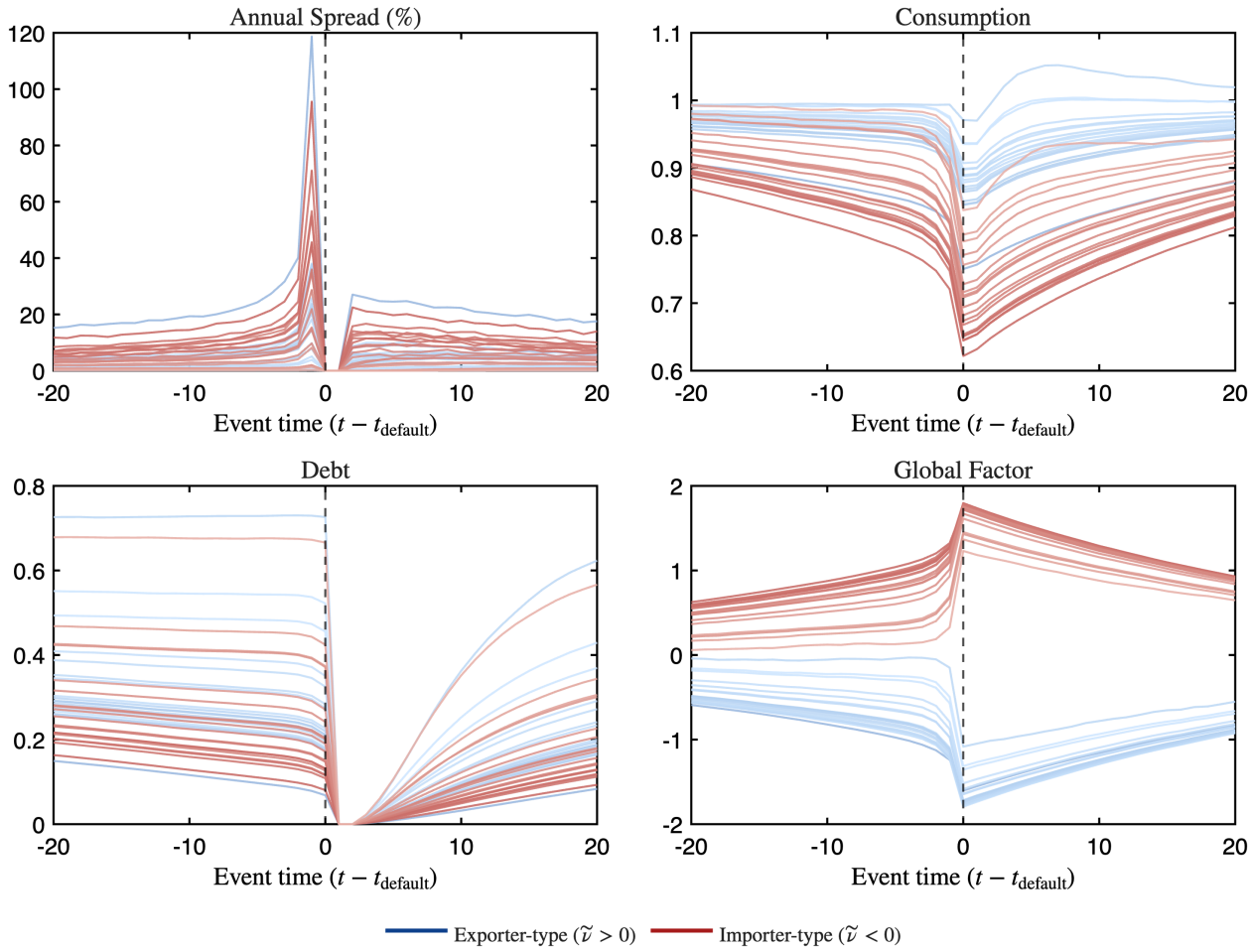
(iv) Opposite patterns of debt comovement. Mean debt, in units of trade income, ranges from 0.26 in Mexico to 0.73 in Uruguay. Its covariance with the global factor is positive in all 18 exporter-type countries and negative in all 19 importer-type countries. Higher global commodity prices are therefore associated with greater indebtedness in exporter-type economies, where borrowing costs fall, and lower indebtedness in importer-type economies, where borrowing costs rise.

Section S5.3 of the Supplemental Material compares simulated debt with observed general government debt. The median within-country coefficient of variation is 0.54 in the model and 0.38 in the data. Debt levels are not directly comparable because model debt represents external borrowing measured relative to commodity income.

F.2 Dynamics Around Default Events for the 37 Countries

Figure 14 shows distinct pre-default dynamics across the 37 calibrated economies. Defaults in exporter-type countries are preceded by negative realizations of the global factor, which weaken trade income, whereas defaults in importer-type countries follow positive realizations, which raise consumption costs. In both groups, these adverse shocks are accompanied by rising sovereign spreads. Consumption reaches a trough around default, while debt falls discontinuously at $\tau = 0$ as repayment is suspended. Countries with greater exposure to the global factor exhibit larger pre-default increases in spreads.

Figure 14: Dynamics Around Default Events – All 37 Sample Countries.



Note: Each line shows a country's simulated average dynamics around default. Blue denotes exporter-type economies and red importer-type economies, with darker shades indicating greater exposure to the global factor. The dashed line marks the default date ($\tau = 0$). Results are based on 500 simulations of 10,000 periods per country.